

Psychology

Emotion Theory

- **James-Lange Theory**: emotional stimuli cause physical response which drives subjective feelings
 - Sensory input → somatic state → subjective feeling
- **Cannon-Bard Theory**: emotional stimuli cause simultaneous physical response and subjective feeling
 - Sensory input → (SIMULTANEOUS) somatic state + subjective feeling
- **Schachter-Singer Two-Factor Theory**: cognitive appraisal and physiological state interact to give rise to subjective feeling
 - Sensory input → somatic state → subjective feeling
 - Sensory input → cognitive appraisal → subjective feeling

Diagnosing Disorders

- **Descriptors**
 - **Provisional**: term used if one presumes that criteria are (or will be) met for the disorder but either 1) patient information is unavailable (and expected to be confirmed upon inquiry) or 2) required symptoms are present, but the duration criteria is not yet (but expected to be) met; term goes after the specific diagnosis that the person is presumed to have
 - <specific diagnosis>, provisional
 - **Unspecified**: term used to diagnose when a patient definitely does not meet criteria for a specific disorder but has significant symptoms within a given diagnostic category; term goes before the name of the diagnostic category (not a specific diagnosis)
 - Unspecified <category name>
 - **"Prior History" & "In Remission"**: used when a patient has been previously diagnosed with a disorder, but the symptoms are no longer present
 - Prior history: implies that the prior disorder will not recur
 - In remission: implies that the prior disorder is currently controlled, but is likely to recur
- **Specifiers**
 - **With melancholic features**: used for depressive episodes characterized by symptoms such as: severe anhedonia, lack of mood reactivity, profound despondency and guilt, depression worse in the morning, early-morning awakenings, significant appetite loss
 - **With atypical features**: used for depressive episodes characterized by symptoms such as: weight increase, sleep increase, mood reactivity, leaden paralysis
 - **With psychotic features**: used for depressive or manic episodes characterized by delusions and/or hallucinations
 - **With mood-congruent psychotic features**: the content of the delusions/hallucinations is consistent with the typical themes of that mood state (depression/mania)
 - **With mood-incongruent psychotic features**: the content of the delusions/hallucinations is inconsistent with the typical themes of that mood state (depression/mania)
 - **With catatonia**: used for depressive or manic episodes characterized by psychomotor symptoms such as: mutism, immobility, waxy flexibility, stereotypies, other odd posturing
 - **With peripartum onset**: used for depressive or bipolar disorders in which the onset of the mood episodes occurs either during pregnancy or within 4 weeks postpartum
 - **With seasonal pattern**: used for depressive or bipolar disorders that show a consistent temporal relationship between time of year and the mood episode
 - **With rapid cycling**: used for bipolar disorders in which 4 or more mood episodes occur per year (mania, hypomania, or depression)

Neurodevelopmental Disorders

- **Disorders**
 - **Intellectual Disability**: deficits in intellectual functioning and deficits in adaptive functioning (limitations in communication, social participation, independent living across multiple environment) that begins before 18 years old
 - Intellectual functioning

- IQ score: at least 2 standard deviations below the mean (score less than or equal to 70 +/- 5)
 - Test: Wechsler intelligence tests
- Adaptive functioning
 - Failure to achieve developmentally appropriate daily living skills in at least one of the following domains → conceptual, social, practical; assessed using a standardized rating scale (vineland adaptive behavior scales)
- Classification:
 - **Mild**: can live independently with minimum levels of support; immature social interactions
 - **Moderate**: varying degrees of support needed such as those available in group homes; limited communication and socialization skills, aware of their limitations
 - **Severe**: requires daily assistance with self-care activities and safety supervision; uses simple speech and gestures, have maladaptive behavior
 - **Profound**: requires 24-hour care; non-verbal communication
- **Global Developmental Delay**: a child less than 5 years old that has not met developmental milestones in several areas of functioning including learning how to walk/talk and learning appropriate social and emotional interactions
 - Notes: clinical severity level cannot be reliably assessed with standardized tests
- **Specific Learning Disorders**: difficulty learning and using academic skills at a level less than expected for age
 - Criteria: having at least 1 symptom for at least 6 months → slow/inaccurate/effortful word reading, difficulty understanding meaning of what is read, difficulty spelling, difficulty with written expression, difficulty with numbers/calculation, difficulty with mathematical reasoning; academic skills deficiency must cause significant interference with academics/occupation/daily living and can be confirmed with standardized tests and clinical assessment
 - Specifiers
 - **Dyslexia**: impairment in reading
 - **Dyscalculia**: impairment in mathematics
 - Impairment in written expression
 - Classification
 - **Mild**: mild difficulties; may learn to function well with initial support services
 - **Moderate**: marked difficulties; needs accommodations/support to complete tasks
 - **Severe**: severe difficulties; needs intensive and individualized teaching
- **Communication Disorders**:
 - **Language Disorder**: persistent difficulty in acquisition and use of language across modalities; there is a decreased vocabulary, limited sentence structure, and impairments in conversing and language ability below expectations for age (typically present before 4 years)
 - **Speech Sound Disorder**: persistent difficulty with speech sound production that interferes with speech intelligibility or prevents verbal communication (speech should be intelligible by age 3-4 and milder cases typically spontaneously recover by 8 years old)
 - **Childhood-Onset Fluency Disorder**: persistent disturbances in the normal fluency and time patterning of speech that are inappropriate for age and level of language skill and the disturbance causes anxiety about speaking or limits effective communication; usually diagnosed by age 6 and may not be present during oral reading, singing, or talking to inanimate objects
 - Symptoms: sound and syllable repetitions, sound prolongations of consonants and vowels, broken words, blocking (pauses in speech), words produced with excessive tension, monosyllabic whole-word repetitions
 - Notes: **stuttering**
 - **Social/Pragmatic Communication Disorder**: persistent difficulties in the social use of verbal and nonverbal communication (deficits in using communication for social purposes,

impairment in ability to change communication to match context, difficulty following the rules of communication, and difficulty understanding implicit meanings) and the deficits result in functional limitations in effective communication that interferes with social, academic, and occupational performance; is rarely diagnosed before ages 4-5

- **Autism Spectrum Disorder**: persistent deficits in social communication and social interaction across multiple contexts and restricted, repetitive patterns of behavior, interests, or activities; typically diagnosed by 2 years old
 - Symptoms
 - Deficits in social-emotional reciprocity (limited back-and-forth conversation, failure to initiate or respond to social interactions, does not share experiences/emotion)
 - Deficits in nonverbal communicative behaviors (limited eye contact, use of gestures, facial expressions and a limited understanding of body language and facial expressions)
 - Deficits in developing, maintaining, and understanding relationships (problems with sharing, imaginative play, making friends)
 - Stereotyped or repetitive motor movements, use of objects, or speech
 - Insistence on sameness, inflexible adherence to routines, or ritualized patterns of verbal or nonverbal behavior
 - Highly restricted, fixated interests that are abnormal in intensity or focus
 - Hyper- or hypo reactivity to sensory input or unusual interest in sensory aspects of the environment
 - Specifiers
 - With/without accompanying intellectual impairment
 - With/without accompanying language impairment
 - Notes: **Asperger's Disorder** → Autism Spectrum Disorder without language impairment and without intellectual impairment
- **Attention-Deficit/Hyperactivity Disorder (ADHD)**: 6 or more specific symptoms of inattention and/or 6 or more specific symptoms of hyperactivity-impulsivity and has persisted for at least 6 months
 - **Inattention**: 6 or more specific symptoms of inattention for at least 6 months
 - Lack of attention to details, difficulty in remaining focused, may seem not to be listening, lack of follow through, difficulty organizing tasks, avoids tasks that require sustained effort, loses things, easily distracted, forgetful
 - **Hyperactivity and Impulsivity**: 6 or more specific symptoms of hyperactivity-impulsivity have persisted for at least 6 months
 - **Hyperactivity**: fidgets/squirms in seat, leaves seat when being seated is expected, runs about inappropriately, unable to remain quiet while at play or leisure, on-the-go/motor-driven, talks excessively
 - **Impulsivity**: blurts out, cannot wait for turn, interrupts/intrudes
- Subtypes
 - **ADHD, Combined Presentation**: inattention + hyperactivity/impulsivity (both criteria have 6+ symptoms for 6+ months)
 - **ADHD, Predominantly Inattentive**: inattention (criteria has 6+ symptoms for 6+ months)
 - **ADHD, Predominantly Hyperactive-Impulsive**: hyperactivity/impulsivity (criteria has 6+ symptoms for 6+ months)
- Treatments
 - Stimulants: **amphetamine (Adderall)**, **methylphenidate (Ritalin)**
 - Increases dopamine and norepinephrine in prefrontal cortex
 - Side effects: appetite loss, insomnia, edginess, GI upset; risk for abuse
 - Non-stimulants: **atomoxetine (Strattera)**, **viloxazine**
 - Selectively inhibit norepinephrine reuptake in prefrontal cortex
 - Used when stimulants fail or if there are contraindications (**viloxazine contraindicated with MAOIs**)

Neurocognitive Disorders

- Disorders in which the core feature is acquired dysfunction in a cognitive domain occurring after early life; decline in at least one cognitive domain
- Diagnostic terms
 - **Major NCD**: significant decline in neurocognition where deficits interfere with independence in daily activities (activities of daily living - ADLs)
 - **Mild NCD**: modest decline in neurocognition where deficits do not interfere with the capacity for independence in daily activities (activities of daily living - ADLs)
- Disorders
 - **Delirium**: a disturbance in the level of consciousness/alertness that is characterized by the factors listed below:
 - Deficits in awareness and attention
 - Awareness is assessed by one's alertness and orientation to the environment
 - Attention is assessed by one's ability to direct, focus, sustain, and shift attention
 - An additional cognitive disturbance
 - Deficits in memory/language/executive functioning
 - Delusions
 - Perceptual disturbance (hallucinations, illusions)
 - Sudden onset of symptoms
 - Usually over hours to a few days
 - Typically fluctuate throughout the day
 - Evidence for a direct physiological cause
 - Medical condition
 - Drug intoxication/withdrawal
 - Toxin exposure
 - Typical clinical presentation
 - Hyperactive (increased psychomotor activity, mood lability, agitation, uncooperative)
 - Hypoactive (reduced psychomotor activity, sluggishness, lethargy approaching stupor)
 - Sleep-wake cycle disturbances
 - Diffuse slowing on EEG (theta/delta waves)
 - Pathology
 - Etiology: fever, dehydration, UTI
 - Deficits: central cholinergic functioning, reticular activating system and its ascending connections
 - Risk factors: non-modifiable (poor health, older age) & modifiable (immobilization, poor sleep, benzo use)
 - Course: symptoms persist until cause is reversed, resolution typically occurs within 3-7 days, amnesia for events during delirium is common, increased risk of mortality and morbidity
 - Treatment: treat the underlying medical condition but manage the associated symptoms with antipsychotics (i.e. risperidone) and benzodiazepines (i.e. diazepam) to treat delirium caused by alcohol withdrawal
 - **Dementia**: multiple and severe cognitive impairments without impaired consciousness that is usually progressive and irreversible and occurs most commonly in the elderly
 - **Alzheimer's Dementia**: significant memory impairment plus impairment in at least 1 other cognitive domain; gradual onset with steadily progressive decline (late 70s onset with 10+ year progression)
 - Early stages: memory deficits (encoding deficits + rapid forgetting), anomia (word-finding difficulties)

- Middle stages: further memory and language decline, visuospatial deficits (getting lost), agnosias (loss of knowledge of a perception), mood changes, personality changes, psychosis
 - **Visual agnosia**: visual stimuli are perceived but not identified
 - **Prosopagnosia**: specific inability to identify familiar faces by sight
 - **Auditory agnosia**: sound perceived but not identified
 - **Anosognosia**: lack of awareness of one's disability
- Late stages: severe deficits in all cognitive domains, global aphasia, motor dysfunction, death from opportunistic infections
- Treatment
 - Established
 - Cholinesterase inhibitors (**donepezil**, **galantamine**, **rivastigmine**)
 - NMDA receptor blockers (**memantine**)
 - **Side effects**: hypotension, GI distress, increased fragility/risk of falls
 - New
 - Antibody drug (**aducanumab**) → decreased production and increased clearance of beta-amyloid
 - **Side effects**: brain bleeding, edema
 - Other
 - Non-pharm methods: sensory rooms
 - Antipsychotics: used off-label, black-box warning of increased mortality risk
 - Anticonvulsants + antidepressants
 - Psychosocial considerations: external mnemonics, assessment/restriction of driving, medic alert-safe return program for wanderers
 - Support for caregivers: risk of **Caregiver Syndrome** from psychological, social, and financial sequelae of caregiving (palliative care and respite care help)
- **Vascular Dementia**: dementia that results from multiple infarcts caused by cerebral vascular disease (**periventricular infarction**)
 - Symptoms: **sudden onset with stepwise progression**; **focal/lateralizing neurological signs**, headaches and seizures are more common in the early stages
 - Treatment: address underlying CVD
- **Frontotemporal Dementia/Pick's Disease (FTD)**: similar to Alzheimer's dementia but has "frontal" signs prominent early in the dementia (**disinhibition**, **personality change**); hypometabolism and lobar atrophy of prefrontal and temporal cortex (has **Pick bodies** in Pick's disease)
- **Lewy Body Dementia (LBD)**: fluctuating cognition/alertness, **visual hallucinations**, mild Parkinsonism; **Lewy bodies** present, REM sleep behavior disorder often precedes onset of LBD by many years
 - Treatment: **cholinesterase inhibitors** (may improve hallucination and agitation), **antipsychotics** (cause serious reactions, are typically avoided), **L-DOPA** (may worsen psychosis, used in low doses)
- **Parkinson's Disease with Dementia (PDD)**: up to 50% of Parkinson patients develop dementia; similar presentation to LBD
 - 1-year rule
 - If dementia develops **12+ months after motor symptoms** → PDD
 - If dementia develops **within 12 months of motor symptoms** → LBD

- **Huntington Disease Dementia (HDD)**: dementia develops after the onset of choreoathetosis and psychiatric symptoms
- **Prion Disease**: dementia progresses rapidly over a few months with death within 2 years; spongiform degeneration (tiny holes) gives brain a “spongy” appearance
- **Reversible Dementias**: due to infections, normal pressure hydrocephalus, etc.
- **Benign Senescent Forgetfulness**: cognitive decline associated with the normal aging process
- **Mild Cognitive Impairment**: cognitive decline exceeding the normal aging process but that does not cause impairment in activities of daily living; are at increased risk for developing dementia
- **Pseudodementia** (Dementia Syndrome of Depression): cognitive deficits associated with a psychiatric disorder such as depression; deficits do not reflect a true dementing process but relate to the effects of the psychiatric condition
- Testing
 - Memory tests
 - **Free recall**: no hints; help to establish a memory deficit exists but do not identify the specific memory process affected
 - **Cued recall**: hints given; help to determine if the memory deficit is encoding-based or retrieval-based
 - **If cues don't help**: encoding didn't occur, thus the problem is encoding-based
 - **If cues help**: encoding did occur, thus the problem is retrieval-based

Anxiety Disorders

- All share a common theme of inappropriate anxiety that becomes abnormal when it is elicited by inappropriate cues (or no cues), excessive in intensity, excessive in duration
 - “Fear” (response to imminent threat) and “anxiety” (response to anticipation of future threat) are used interchangeably
- Disorders
 - **Panic disorder**: intense fear with somatic or cognitive symptoms
 - Criteria: recurrent and unexpected panic attacks (unspecified #)
 - Shows: fear of future panic attacks AND/OR a significant change in behavior related to the attacks (i.e. avoidance of exertion)
 - Symptoms: cardiac symptoms, sweating, trembling, shortness of breath, choking, chest pain, gastrointestinal symptoms, feeling faint, feeling detached, fear of losing control, fear of dying, tingling/numbness, chills/hot flashes; abrupt onset with episode peak within minutes
 - Trigger: **unexpected**
 - **Agoraphobia**: fear of situations due to fear of inability to escape/seek help
 - Criteria: >2 of the following situations are feared/avoided
 - Using public transportation, being in open spaces, being in closed spaces, standing in line or being in a crowd, being outside of the home alone
 - These situations are feared/avoided because of thoughts that escape might not be difficult or help might not be available if symptoms arise
 - Notes: agoraphobia can often develop in panic disorder patients, but can be mutually exclusive
 - Trigger: being in multiple “difficult to escape/rescue” situations because something unwanted could happen
 - **Specific phobia**: excessive, persistent, and disproportional fear of an object or situation
 - Notes: fear invariably occurs when exposed to the stimulus, or the person avoids the phobic situation; the fear causes marked distress/impairment and is not accounted for by a different disorder
 - Subtypes: animal, natural environment, situational, blood-injection injury (common for patients to faint), other
 - Trigger: **a specific object/situation (but not social/performance setting)**

- **Social anxiety disorder**: excessive, persistent, and unrealistic fear of social situations involving possible scrutiny by others due to fear of negative evaluation
 - Notes: social situations are avoided or endured with great distress
 - Specific "performance only" anxiety is diagnosed if the fear is restricted to performing in public (i.e. public speaking, athletics, etc.)
 - Trigger: **a social/performance setting**
- **Generalized anxiety disorder**: persistent uncontrolled anxiety about multiple events
 - Symptoms: restless/on edge, decreased concentration, muscle tension, fatigued, irritability, insomnia
 - Trigger: **everything (person constantly worries)**
- **Separation anxiety disorder**: developmentally inappropriate and excessive anxiety concerning separation from major attachment figures
 - Symptoms: person experiences marked distress (mental and physical) about anticipated/actual separation, losing or harm befalling a major attachment figure or self that leads to separation, going out (i.e. school work, etc.) because of separation fears
 - Trigger: **separation from attachment figure**
- **Selective mutism**: refusal to speak in specific situations despite fluent speech in other contexts
 - Notes: written communication and social play may also be affected, refusal to speak is anxiety-based; not due to a lack of language knowledge or a communication disorder (i.e. stuttering)
 - Trigger: **speaking (manifested by mutism)**
- Development of anxiety: often learned through classical conditioning (i.e. specific phobia) via learning through paired association
- Treatments: treated by diminishing emotional responsiveness
 - **Cognitive-Behavioral Therapy (CBT)**
 - Modifying one's thoughts to change emotion (cognitive)
 - Using exposure techniques to extinguish the emotion (behavioral) - like systematic desensitization, exposure therapy, etc.
 - **Anxiolytic Medications**
 - Antidepressants
 - SSRIs (i.e. paroxetine, sertraline)
 - Benzodiazepines
 - GABA agonists (i.e. diazepam, alprazolam)
 - Recommended only for short-term use due to addiction potential

Obsessive-Compulsive & Related Disorders

- Behaviors must cause distress/impairment & are not better explained by another disorder
- Disorders
 - **Body Dysmorphic Disorder (BDD)**: preoccupation with a perceived flaw in physical appearance (flaw is minimal or non-observable)
 - Criteria: repetitive behaviors or mental acts are performed in response to the appearance concerns; preoccupation is not better accounted for by an eating disorder and must cause functional impairment (i.e. sequestering oneself) otherwise, the preoccupation may be considered normal vanity
 - Notes: the belief may be of delusional intensity (i.e. BDD with absent insight [delusional beliefs])
 - **Hoarding Disorder (HD)**: accumulation of possessions in living areas that compromises their intended use
 - **Excoriation Disorder**: recurrent, unwanted skin picking that causes lesions
 - **Trichotillomania**: recurrent, unwanted pulling out of one's hair
 - **Obsessive-Compulsive Disorder (OCD)**: recurrent obsessions and/or compulsions
 - **Obsessions**: intrusive recurrent thoughts, urges, or images that increase distress (i.e. fear of contamination)

- **Compulsions**: repetitive behaviors/mental acts that are performed to decrease distress (i.e. washing)
- **Criteria**: compulsions may not always be logically related to the obsession, obsessions/compulsions are time consuming or distressful/disruptive, symptoms are not explained by another disorder (i.e. weight preoccupation in eating disorders)
- **Notes**: the obsessional belief can be of delusional intensity (100% certainty) and is denoted using the following specifier: OCD "with absent insight (delusional beliefs)"
- **Treatments**: **behavioral therapy** (exposure and response prevention), antidepressants (SSRIs), **neurosurgery/neuro radiation** (create lesions to anterior cingulate gyrus or anterior limb of the internal capsule), **deep brain stimulation**

Feeding & Eating Disorders

- **Purge**: use of vomiting, laxatives, diuretics, or enemas to eliminate calories
- **Binge**: eating a large amount within a discrete period; eating is out of control during the episode
- **Disorders**
 - **Anorexia Nervosa**: restriction of food that leads to a person being **significantly underweight (BMI < 18.5)**, having an intense fear of gaining weight, and experiencing body image disturbance
 - **BMI Stages**
 - Mild: 18.49-17
 - Moderate: 16.99-16
 - Severe: 15.99-15
 - Extreme: < 15
 - **Types**
 - **Binge-Eating/Purging Type**: if the person **recurrently binges and/or purges**
 - Symptoms: calloused knuckles (Russell's sign), dental enamel erosion, salivary gland inflammation (parotid gland hypertrophy, chipmunk cheeks), subconjunctival hemorrhage, hypochloremia, hypokalemia
 - **Restricting Type**: if the person does not recurrently binge or purge; weight loss is maintained through **dieting, fasting, and/or over-exercising**
 - Symptoms: amenorrhea, decreased sex drive, constipation, hair loss, low BMI, hypothermia, bradycardia, hypotension, **lanugo** (soft, fine, baby-like hair), dry skin, brittle nails, anemia, leukopenia, low bone mineral density (osteopenia + increased risk of fractures), **hypercholesterolemia**
 - Notes: 90% are female; treatment is typically initiated by family, not by the patient
 - Treatment: in-patient hospitalization, **cognitive-behavioral therapy**
 - **Bulimia Nervosa**: recurrent binge-eating & inappropriate compensatory behavior for bingeing (purging & non-purging) that must occur for more than once a week for at least 3 months; self-evaluation is unduly influenced by body shape
 - Symptoms: BMI in normal/slightly overweight range, potentially **low serotonin levels**, calloused knuckles (Russell's sign), dental enamel erosion, salivary gland inflammation (parotid gland hypertrophy, chipmunk cheeks), subconjunctival hemorrhage, hypochloremia, hypokalemia
 - Notes: no subtypes of this disorder, 90% are female; patient typically initiates treatment (disorder is distressful to the patient)
 - Treatment: out-patient treatment, **cognitive-behavioral therapy, fluoxetine**
 - **Binge-Eating Disorder**: binge eating associated with several features like rapid eating, eating until uncomfortably full, eating when not physically hungry, eating alone due to shame about quantity, feeling disgusted/guilty/depressed after a binge **without** any inappropriate compensatory behaviors; behavior must occur more than once a week for 3 months
 - Symptoms: BMI is in obese range but high BMI is not required for diagnosis
 - Treatment: **cognitive-behavioral therapy, antidepressants, stimulants (lisdexamfetamine)**
 - **Unspecified Eating Disorder**: patients do not meet the criteria for either Anorexia Nervosa, Bulimia Nervosa, or Binge-Eating Disorder, but still have **subclinical eating problems**
- **SCOFF Questions**: used to indicate likely eating disorder (2 or more is indicative)

- Sick: do you make yourself sick because you feel uncomfortably full?
- Control: do you worry you have lost control over how much you eat?
- One: have you recently lost more than one stone (14 lbs) in a 3 month period?
- Fat: do you believe yourself to be fat when others say you are too thin?
- Food: would you say that food dominates your life?

Somatic Symptom & Related Disorders

- **Somatic Symptom Disorder (SSD)**: abnormal behaviors/thoughts/feelings in response to certain distressing somatic symptoms
 - Criteria: 1 or more distressing/disruptive somatic symptoms; at least 1 indicator of excessive thoughts/feelings/behaviors about the symptoms such as: disproportionate thoughts about the seriousness of the symptom, high levels of anxiety about the symptom or health, excessive time/energy devoted to the symptom
 - Notes: patient has a distressing physical complaint with "excessiveness" in response to that distressing physical complaint
 - Differential: Delusional Disorder (DD), Somatic Type: a Schizophrenia Spectrum Disorder characterized by a persistent fixed, false belief about body/health (the belief is held with delusional intensity [100% certainty]) whereas in SSD and IAD, the belief is less-strongly held
- **Illness Anxiety Disorder (IAD)**: preoccupation with having/acquiring a serious illness
 - Criteria: somatic symptoms are not present, or, if present, the symptoms are mild (i.e. normal physiological sensations, benign/self-limiting dysfunction, bodily discomfort that is not usually indicative of disease, etc.); patient performs excessive health-related behaviors or shows maladaptive avoidance
 - Notes: patient does not have a distressing physical complaint but nonetheless worries about one's health
 - Differential: Delusional Disorder (DD), Somatic Type: a Schizophrenia Spectrum Disorder characterized by a persistent fixed, false belief about body/health (the belief is held with delusional intensity [100% certainty]) whereas in SSD and IAD, the belief is less-strongly held
- **Conversion Disorder**: altered voluntary motor or sensory function
 - Criteria: evidence of incompatibility of the symptom with the presenting neurological disease
 - Subtypes: weakness/paralysis, abnormal movement, swallowing symptoms, speech symptoms, attacks/seizures, anesthesia/sensory loss, special sensory symptoms (i.e. visual), mixed symptoms
- **Factitious Disorder**: a person fakes/induces (feigns) physical or psychological symptoms, in themselves or others, in the absence of obvious "external" rewards
 - Notes: satisfaction is from gaining medical attention and being in the patient/sick role (primary gain); unconscious motivation
 - Specifiers
 - **Factitious Disorder Imposed on Self (Munchausen's Syndrome)**: person feigns symptoms in themselves
 - **Factitious Disorder Imposed on Another (Munchausen's Syndrome, by Proxy)**: person feigns symptoms in another individual
 - Differential
 - **Malingering**: a person fakes/induces (feigns) physical or psychological symptoms, in themselves or others, for "external" rewards (i.e. avoiding work)
 - Notes: known as secondary gain; conscious motivation, complaints cease after gaining reward
 - Specifiers
 - **Malingering**: person feigns symptoms in themselves
 - **Malingering, by Proxy**: person feigns symptoms in another individual

Trauma & Stressor-Related Disorders

- Psychological distress in response to a stressful event
 - Traumatic stressor: involves actual/threatened death or serious injury
 - Non-traumatic stressor: perception that environmental demands exceed one's resources to cope

- Disorders
 - **Post-Traumatic Stress Disorder (PTSD)**: exposure to a traumatic stressor (actual or threatened death/serious injury/sexual violence)
 - Notes: exposure may include direct experience of the event, witnessing event, or learning about an event that happened to a close family member/friend
 - Symptoms: 1+ symptoms from each of the four categories must be present
 - **Intrusive symptoms**: dreams, automatic recollections, feeling like the event is recurring, psychological or physiological distress when encountering symptoms that remind the patient of the event
 - **Avoidance symptoms**: individual avoids thoughts, places, or conversations that remind them of the event
 - **Negative alterations in cognition and mood**: negative beliefs/expectations, negative emotional states, inability to experience positive emotion, diminished interest/participation in activities, detachment/estrangement from others, inability to remember an important aspect of the traumatic event
 - **Alterations in arousal and reactivity**: sleep disturbance, irritable and angry outbursts, reckless/self-destructive behavior, concentration problems, hypervigilance, exaggerated startle response
 - Criteria: symptoms must last longer than 1 month and must usually begin within 3 months of the trauma (but can begin at any time)
 - Treatments: psychotherapy/behavioral therapy, antidepressants (SSRIs)
 - **Acute Stress Disorder (ASD)**: exposure to a traumatic stressor but for a more brief duration than PTSD
 - Symptoms: numerous PTSD-like symptoms must develop from the four PTSD symptom categories or an additional category (i.e. dissociative symptoms)
 - Altered sense of one's surroundings or oneself
 - Inability to remember an important aspect of the traumatic event(s), typically due to dissociative amnesia
 - Criteria: duration of symptoms is 3 days to 1 month after trauma exposure
 - Treatments: psychotherapy/behavioral therapy, antidepressants (SSRIs)
 - **Adjustment Disorder**: significant and disproportional emotional/behavioral symptoms in direct response to a stressor (can be ordinary or traumatic)
 - Notes: Has acute onset and brief duration; symptoms cannot be explained by another disorder
 - Subtypes
 - With depressed mood, with anxiety, with disturbance of conduct (i.e. vandalism), with mixed anxiety and depressed mood, with mixed disturbances of emotion & conduct, unspecified
 - Treatments: psychotherapy, support groups, psychotropic medications, hospitalization
 - **Reactive Attachment Disorder (RAD)**: consistent pattern of inhibited, emotionally withdrawn behavior
 - Symptoms: at least 2 persistent social and emotional disturbances (minimal social and emotional responsiveness to others, limited positive affect, unexplained irritability/sadness/fearfulness during nonthreatening interactions), at least 1 extreme of insufficient childcare (persistent lack of emotional needs for comfort/stimulation/affection, repeated changes of primary caregivers, rearing in unusual settings that limits attachment), insufficient childcare precedes withdrawn behavior
 - Etiology: direct result of pathogenic care in early childhood (under 5 years old) like social emotional neglect or repeated changes in the adult caregiver
 - **Disinhibited Social Engagement Disorder (DSED)**: similar to RAD with changes in externalizing vs internalizing behaviors
 - Symptoms: child approaches and interacts with unfamiliar adults in 2+ ways (reduced/absent reticence in interacting with unfamiliar adults, age and culturally inappropriate verbal or physical behavior, diminished or absent checking back with adult

- caregiver after venturing away - even in unfamiliar settings, willingness to go off with an unfamiliar adult)
- Etiology: direct result of pathogenic care in early childhood (under 5 years old) like social emotional neglect or repeated changes in the adult caregiver
- **Persistent Grief Disorder (PGD)**: 1+ months of clinically significant bereavement symptoms almost daily that occurs 12+ months after the death of a close person
 - Symptoms: intense longing and/or preoccupation with thoughts of the deceased person and multiple additional symptoms related to the death such as: identity disruption, disbelief about the death, avoidance of reminders about the death, intense emotional pain, difficulty re-engaging in activities, emotional numbness, feeling of life as meaningless, intense loneliness
 - Notes: symptoms must not be explained by major depressive disorder or another mental disorder; if patient is a child/adolescent, PGD can be diagnosed after at least 6+ months have elapsed since the death of a close person

Dissociative Disorders

- **Dissociation**: splitting off from conscious awareness an aspect of cognition that helps to create a sense of self-identity; a “last-ditch” effort to respond to overwhelmed coping mechanisms; primarily a dysfunction of the process of retrieving memories
- Disorders
 - **Dissociative Amnesia**: memory loss for autobiographical information not caused by another disorder
 - Types of Amnesia
 - **Localized amnesia**: total loss of personal memory during a circumscribed period of time
 - **Selective amnesia**: some limited recall of personal memories during a circumscribed period of time
 - **Generalized amnesia**: loss of personal memory of entire life up to and including the triggering event
 - **Biological amnesia**: patient will have difficulty learning new information (anterograde memory loss) in addition to past memory loss
 - **Psychological (dissociative) amnesia**: patient learns new information well, only past memory loss (retrograde memory loss) will be present
 - Specifiers: fugue/dissociative **fugue** (purposeful travel or bewildered wandering associated with amnesia for identity or other autobiographical information)
 - **Dissociative Identity Disorder (DID)**: disruption of individual identity characterized by 2+ distinct personality states (the Primary/Host & the Alter)
 - Symptoms: inability to recall personal information (as evidenced by frequent memory gaps in the Primary personality while an Alter takes control)
 - **Depersonalization/Derealization Disorder**: experiencing either/both of depersonalization/derealization phenomena
 - **Depersonalization**: experiences of unreality, detachment or being an outside observer of one's thoughts, feelings, sensations, body or actions
 - **Derealization**: experiences of unreality or detachment with respect to surroundings (i.e. objects and/or environment seem unreal or dreamlike)
 - Criteria: reality testing remains intact (patient knows the perceptual experience is just a misperception); symptoms result in functional impairment, symptoms are not due to physiological effects of a substance or another medical condition, symptoms are not better explained by another mental disorder
- Treatments: **psychotherapy**, hypnosis

Mood

- Manic ↔ hypomanic ↔ euthymic ↔ dysthymic ↔ depressed

Depressive Disorders

- Characterized as low mood without abnormally high mood periods (ranging from dysthymic $\longleftrightarrow$ depressed)
- Disorders
 - **Persistent Depressive Disorder (PDD)**: chronic depressed mood for 2 or more years
 - Forms: long-lasting major depressive episode (MDE) or dysthymia
 - **Dysthymia**: chronic depressed mood plus 2 or more of the following symptoms:
 - Poor appetite or overeating, insomnia/hypersomnia, low energy, low self-esteem, trouble concentrating or making decisions, feelings of hopelessness
 - Specifiers:
 - **PDD with persistent MDE**: depression is characterized by a prolonged MDE
 - **PDD with pure dysthymic syndrome**: depression is characterized by prolonged dysthymia but no MDE occurs
 - **PDD with intermittent MDEs**: depression is characterized by dysthymic periods and MDEs
 - Treatment:
 - The same treatment for MDD is used for the dysthymic subtype of PDD; **brain stimulation therapies** may also be effective with treatment-resistant individuals
 - **Premenstrual Dysphoric Disorder (PMDD)**: dysphoric symptoms surrounding a woman's menstrual cycle
 - Criteria: symptoms must present in the week before menses onset, must improve a few days after menses onset, and must minimize in the week post-menses
 - Symptoms: mood lability, irritability, dysphoria, anxiety, anhedonia, problems concentrating, lethargy, appetite and sleep changes, and other physical symptoms (i.e. breast tenderness, weight gain, etc.); symptoms must cause clinically significant distress or functional impairment
 - Treatment: **SSRIs**
 - **Disruptive Mood Dysregulation Disorder (DMDD)**: severe temper outbursts in children under 18 for at least x3 episodes/week
 - Criteria: child must be at least 6 years old, symptoms must begin before 10 years old, and it cannot be diagnosed after 18 years old
 - Symptoms: sad/irritable/angry mood almost daily, reaction is disproportionate to situation, symptoms are present in multiple settings
 - **Major Depressive Disorder (MDD) / Unipolar Depression**: individual must experience 1+ major depressive episode (MDE) for diagnosis; cannot have history of mania or hypomania
 - Criteria: at least 2 weeks of 5+ of the following symptoms, 1 of which must be affective
 - Affective: depressed mood, anhedonia
 - Neurovegetative: change in appetite/significant weight change, insomnia/hypersomnia, loss of energy
 - Cognitive: psychomotor changes, feeling guilty/worthless, decreased concentration, thoughts of death/suicidal ideation
 - Symptoms
 - **Sleep disturbances**: insomnia (initial insomnia, night waking/middle insomnia, early morning waking/terminal insomnia), hypersomnia, altered sleep architecture (decreased REM latency, increased REM time, decreased SWS/N3)

Bipolar Disorders

- Disorders in which high mood is the common theme (typically ranging from manic $\longleftrightarrow$ hypomanic but also includes dysthymic $\longleftrightarrow$ depressed)
- Disorders
 - **Bipolar I Disorder (BPI)**: individuals must experience at least one manic episode
 - Criteria: symptoms must last for at least 1+ week (or any duration if hospitalization is involved), must cause marked impairment in functioning
 - Notes: most BPI patients also experience MDEs, but they are not necessary for diagnosis

- **Mania**: abnormally elevated mood or irritability and increased energy; 3+ additional symptoms required
 - Inflated self-esteem/grandiosity, decreased need for sleep, pressured speech, flight of ideas, distractibility, increased goal-directed activity/psychomotor agitation, excessive involvement in “risky” activities
- **Bipolar II Disorder (BPII)**: at least 1 MDE and at least 1 hypomanic episode
 - Criteria: change in mood must be “unequivocal” and “uncharacteristic”; hypomanic episode must last 4+ days
 - **Hypomania**: same symptoms as mania but of a different severity (mania is marked impairment in functioning and can involve psychosis while hypomania is not characterized by either of those two things)
- **Cyclothymic Disorder**: periods of hypomanic symptoms that fluctuate with periods of depressive symptoms for 2+ years
 - Criteria: never met criteria for a MDE or a manic episode
 - Symptoms: similar to BPI but with less extreme mood shifts

Sex & Gender-Related Disorders

- **Gender Dysphoria (GD)**: incongruence between one’s assigned (natal) gender and one’s experienced gender, resulting in clinically significant distress or functional impairment
 - Symptoms: belief that one is the experienced gender, desire to act and be treated as the experienced gender, desire to be rid of one’s assigned sex characteristics and to possess those of the experienced gender, conviction that one has the typical feelings/reactions of the experienced gender
- Sexual Dysfunctions: problems related to the normal sexual response cycle which are caused primarily by psychological factors; a medical condition may co-exist, but is insufficient to explain the sexual dysfunction
 - Disorders
 - **Female Sexual Interest/Arousal Disorder**: problems with libido and/or physiological arousal
 - **Female Orgasmic Disorder**: a delay in, inability to, or low-intensity orgasm(s)
 - **Genito-Pelvic Pain/Penetration Disorder**: difficulties with vaginal penetration or vulvovaginal/pelvic pain during intercourse/penetration attempts
 - **Male Hypoactive Sexual Desire Disorder**: problems with low libido
 - **Erectile Disorder**: problems obtaining or maintaining an erection
 - **Premature Ejaculation**: ejaculation occurring during partnered activities within ~ 1 minute following vaginal penetration
 - **Delayed Ejaculation**: marked delay in, or absence of, ejaculation
 - Treatments
 - Lack of Sexual Desire
 - **Dual Sex Therapy (Sensate Focus Exercises)**: patient and partner complete prescribed “homework assignments” initially involving non-sex activities and gradually progressing to sexual activities over time. When the focus is placed on pleasurable sensations without the pressure to perform, natural sexual responsiveness occurs
 - Premature Ejaculation
 - **Squeeze Technique**: partner squeezes the glans penis to produce discomfort and causes the erection to subside
 - **Start and Stop Technique**: partner ceases stimulation to reduce pleasure and causes the erection to subside
 - **Antidepressants**: SSRIs - a common side effect of antidepressants is a disruption in sexual desire and reduced responsiveness
- **Paraphilic Disorders**: intense and persistent deviant sexual interests as manifested by fantasies, urges, or behaviors
 - Criteria: acts on the urge with a non-consenting person, or acts on the urge with a consenting person (or acts on an urge unrelated to people) and the actions cause significant distress/functional

impairment, or does not act on the urge but the urge causes significant distress and/or functional impairment

- Etiology: classical conditioning may play a role in some of the deviant behavior seen in these disorders
- Disorders
 - **Exhibitionistic Disorder**: exposing one's genitals to an unsuspecting stranger
 - **Voyeuristic Disorder**: observing unsuspecting persons either naked, in the process of disrobing, or engaging in sexual activity (person must be at least 18 years old)
 - **Fetishistic Disorder**: non-living objects or a highly-specific sexual focus on a non-genital body part
 - **Transvestic Disorder**: cross-dressing (wearing clothing of the opposite sex)
 - **Frotteuristic Disorder**: touching or rubbing against a non-consenting person
 - **Sexual Sadism Disorder**: causing psychological/physical suffering to another person
 - **Sexual Masochism Disorder**: receiving psychological/physical suffering
 - **Pedophilic Disorder**: sexual activity between a prepubescent child (13 years old or younger) and a perpetrator (16 years old or older) - perpetrator must be 5 or more years older than the child
 - **Unspecified Paraphilic Disorder**: a stimulus/situation that is not captured by the previous diagnoses
- Treatments
 - Behavioral Therapy
 - **Aversion Therapy**: while engaging in (or thinking about) one's paraphilia, an actual noxious stimulus is applied to the patient
 - **Covert Sensitization**: while fantasizing about one's paraphilia, a patient imagines an aversive consequence
 - Medication
 - **Antidepressants**: SSRIs (reduce libido, control impulses, reduce obsessions)
 - **Anti-Androgens**: "chemical castration"; Medroxyprogesterone acetate (reduce testosterone, diminish sex drive and thus deviancy)

Schizophrenia Spectrum and Other Psychotic Disorders

- **Psychotic**: implies a break from reality in thinking and/or perceptions
 - Psychotic symptom domains
 - **Delusions**: core; a fixed false belief that is not amenable to change, even with conflicting evidence (delusion of grandeur, persecutory, somatic, Capgras delusion → someone replaced by an imposter)
 - **Delusion of control**: external force controlling one's behavior
 - **Delusion of reference**: belief that an external action (i.e. television) refers to or has special meaning to oneself
 - **Thought broadcasting delusions**: thoughts being transmitted to others
 - **Hallucinations**: core; a sensory perception without an external stimulus (auditory, single voice, voices conversing, running commentaries, command hallucinations, derogatory)
 - **Disorganized Speech/Thinking**: core; speech (thinking) lacks the normal, logical connections between words or thoughts - patients with this symptom are described as having a formal thought disorder
 - **Derailment/loose association**: speech with unrelated/remotely-related connections between thoughts
 - **Clang association**: speech with word association based on rhyming
 - **Word salad**: speech with no meaningful relationship between words within a sentence
 - **Disorganized Behavior/Catatonia**:
 - **Disorganized**: non-goal-oriented behavior (i.e. unable to take care of activities of daily living, grossly inappropriate mood/affect)

- **Catatonia**: multiple motor/behavioral abnormalities that reflect diminished reactivity to the environment (posturing, waxy flexibilities, stereotypies, non-responsiveness, mutism)
 - **Negative Symptoms**: diminished emotional expression, avolition, alogia/poverty of speech, anhedonia, asociality
- Symptom types
 - **Negative symptoms** (domain 5): thoughts, behaviors, or perceptions that normally exist that are now absent or markedly diminished
 - **Positive symptoms** (domains 1-4): thoughts, behaviors, or perceptions that are distorted or in excess of normal function
- Disorders
 - **Schizophrenia**: an active phase for 1 month or longer defined as at least 2 psychotic domain symptoms with at least 1 being a core symptom, and continuous signs of the disturbance for 6 months or longer (prodromal + active + residual phases)
 - Etiology: a "lesion" occurs during early brain development; effects of the lesion are seen as brain undergoes continued maturation
 - Evidence: hippocampal cellular disorganization, increased neurological soft-signs, increased minor physical anomalies
 - L/R confusion, agrophesthesia, astereognosis, curved 5th finger
 - Onset
 - Typical onset: late teens with slightly later onset for women
 - Late onset: older than 40 years old; most likely a female with positive symptoms and a less severe course
 - Childhood onset: younger than 13 years old; severe form of disorder with more deficits in neurodevelopment
 - Associated features: low socioeconomic attainment (downward drift), nicotine addiction, suicide risk and aggressive tendencies (if patient experiences command hallucinations or if the patient feels threatened due to paranoid delusions)
 - Progression & outcome: chronic with slow gradual decline or fluctuating symptoms; moderate to severe disability with episodic hospitalization
 - Prognosis
 - Better: older age of onset, rapid onset, positive symptoms, absent CNS changes, present mood symptoms, female gender
 - Worse: younger age of onset, slow onset, negative symptoms, present CNS changes, absent mood symptoms, male gender
 - **Schizophreniform Disorder**: identical symptoms and diagnostic criteria to schizophrenia but lasts between 1-6 months (sometimes provisional for schizophrenia until 6 month symptom duration is reached)
 - Symptoms: delusions, hallucination, disorganized speech, disorganized or catatonic behavior, negative symptoms
 - **Brief Psychotic Disorder**: at least 1 core psychotic symptom with episode lasting less than 1 month with return to full premorbid functioning
 - Specifier
 - Specify trigger of episode (with/without marked stressor)
 - **Delusional Disorder**: the presence of a delusion that lasts for 1+ months without any other psychotic domain symptoms or functional impairments (aside from the direct impact of the delusion)
 - Specifier
 - **Non-bizarre delusion**: a belief that involves something that could occur in real life (i.e. is physically possible)
 - **Bizarre delusion**: a belief that is clearly implausible and not derived from ordinary life experiences
 - Subtypes
 - **Persecutory type**: belief of malevolent treatment

- **Grandiose type**: belief of having some great (but unrecognized) talent or insight or having made some important discovery
- **Erotomaniac type**: belief that someone else is in love with the individual
- **Jealous type**: belief of infidelity by partner
- **Somatic type**: belief involving bodily functions/sensations (i.e. parasitosis, malodorous, organ dysfunction)
- **Unspecified type**: theme is none of the above
- Notes: a delusion may develop in the context of a close relationship with a psychotic individual → a partner comes to believe the same delusion as the psychotic disorder
- **Schizoaffective Disorder**: active phase of schizophrenia occurs concurrently with a major mood episode (MDE or manic episode) or active phase of schizophrenia that lasts 2+ weeks without mood symptoms → mood symptoms must be present for the majority of the total duration of symptomatic periods of the illness
 - Subtypes
 - **Bipolar Type** (if mania is experienced)
 - **Depressive Type** (if depression is experienced without mania())
- Treatments
 - Antipsychotic drugs
 - 1st generation neuroleptics (D2 antagonists): chlorpromazine, haloperidol
 - Relieves positive symptoms
 - Side effects: extrapyramidal syndrome, neuroleptic malignant syndrome
 - **Extrapyramidal Syndrome**: dystonia (sustained muscle contractions of the head + neck that often develops in the 1st week), parkinsonism (tremor and shuffling that often develops in the first few months), akathisia (excessive movements due to an inner restlessness that often develops in the first few months), tardive dyskinesia (repetitive, involuntary, purposeless movements of the face and extremities that develops as a long-term side effect after years of medication)
 - **Neuroleptic Malignant Syndrome (NMS)**: characterized by tachycardia, hypertension, rapid respiration, fever, extreme rigidity, delirium, and death that usually starts within the first 2 weeks of starting medication or increasing dosage and likely results from a precipitous drop in dopamine that impacts hypothalamic functioning
 - 2nd generation neuroleptics (D2, 3, 4 and 5-HT2A receptor antagonists): clozapine, risperidone, olanzapine
 - Relieves positive and negative symptoms
 - Side effects: agranulocytosis (only clozapine); lowered risk of extrapyramidal syndrome
 - General side effects: weight gain, sedation, sexual dysfunction
 - Psychosocial environmental support: housing and social support, vocational rehabilitation, individual psychotherapy, family education
 - **Electroconvulsive Therapy Augmentation**: use for medication-refractory schizophrenia (i.e. catatonic-type)

Disruptive, Impulse-Control, and Conduct Disorders

- These disorders involve problems with self-control of emotions and specific actions; self-control problems typically violate the rights of others and/or brings the individual into conflict with societal norms or authority figures
- Disorders
 - **Oppositional Defiant Disorder (ODD)**: a frequent and persistent pattern of behaviors/emotions characterized by episodes of angry/irritable mood, argumentative/defiant behavior, and vindictiveness
 - Notes: behaviors do not result in a serious violation of the rights of others; diagnosis can be made at any age but is usually done in childhood

- **Conduct Disorder (CD)**: repeated and persistent serious violation of rights/societal norms with at least 3 symptoms occurring in or across any of the below 4 categories
 - Symptoms
 - Aggressive conduct: bullies, uses weapons, cruelty to people/animals, rape, stolen with confrontation
 - Deliberate property destruction: by fire or other means
 - Deceit or theft: broken in, lies, stolen without confronting
 - Serious violation of rules: breaking curfew, running away, truancy
 - Subtypes
 - Childhood-onset type (< 10 years old): usually boys, characterized by aggressive conduct, often have a history of ODD and are at higher risk of ASPD
 - Adolescent-onset type (10+ years old): less of a bias towards boys, characterized by serious violation of rules, not aggressive conduct
 - Notes: can be diagnosed at any age
- **Antisocial Personality Disorder**: *see description in personality disorders section*
- **Intermittent Explosive Disorder**: recurrent outbursts representing a failure to control aggressive impulses as manifested by either verbal aggression or non-damaging physical aggression occurring frequently, or damaging physical aggression occurring infrequently
 - **Aggression**: grossly disproportionate to stressor, impulsive and/or anger-based, not explained by other causes/disorders
 - Typical description of act: described as brief spells that are precipitated by feeling frustrated followed by self-reproach afterwards; has societal, occupational, and legal consequences
- **Kleptomania**: recurrent, irresistible stealing of unneeded objects (stolen items are unwanted and typically of low value; they are hoarded, given away, or returned)
 - Symptoms: increasing tension before stealing, pleasure/gratification/relief when stealing; no other cause or motivating factor for theft
 - **Differential**: in ordinary shoplifting, the object is the goal. In kleptomania, reducing the tension is the goal, the object itself is typically of no importance
- **Pyromania**: multiple episodes of deliberate fire setting with preceding tension or emotional arousal
 - Symptoms: fascination with fire and fire paraphernalia, pleasure/gratification/relief when setting fires or witnessing/participating in the aftermath; no other cause or motivating factor
 - **Differential**: assess motivation for fire-setting behavior - with arson, it is usually as a statement or as a method of retaliation
- Impulse Control Problems
 - Serotonin: decreased levels associated with poor prefrontal impulse control
 - Dopamine: increased levels associated with reward sensation during impulsive act (stimulates brain's reward pathway) - ventral tegmental area (VTA) to nucleus accumbens
- Treatments
 - Behavioral therapy techniques (avoiding triggers, aversion therapy, exposure & response prevention)
 - Medication: SSRIs (variable success)

Personality Disorders

- Description: pattern of perceiving, relating, and thinking about the environment and oneself that is chronic and pervasive, culturally deviant, maladaptive or distressful
- Onset: in childhood-adolescence with behaviors solidifying with age (typically undiagnosed until adulthood)
- Disorders
 - **Cluster A** (odd, eccentric)
 - **Paranoid Personality Disorder (PPD)**: suspicion that others are being exploitative or deceptive which leads to unjustified doubts about loyalty/trustworthiness, a reluctance to confide, injecting hidden threatening meanings into innocuous remarks, reacting angrily to perceived insults

- Theme: distrust + suspiciousness
- Schizoid Personality Disorder: lack of desire for close relationships/intimacy, lacking close friends, preferring solitary activities, deriving pleasure from very few activities, indifference to praise/criticism, having an emotional coldness/detachment/flat affect
 - Theme: interpersonal detachment
- Schizotypal Personality Disorder: a disorder affecting a variety of areas related to thoughts, perceptions, and behaviors
 - Symptoms
 - Thoughts: odd thinking and speech (vague/metaphorical), odd beliefs or magical thinking, suspiciousness, ideas of reference
 - Perceptions: unusual perceptual experiences
 - Behaviors: odd behavior/appearance (including affect), problematic social relatedness (lack of close friends, excessive social anxiety)
 - Notes: schizotypal personality does not involve psychotic symptoms (i.e. delusions, hallucination, etc.) but these may develop leading to → premorbid personality of schizophrenia
 - Theme: eccentric thoughts, perceptions, and behaviors
- Cluster B (dramatic, emotional, erratic)
 - Histrionic Personality Disorder: needs to be the center of attention, exaggerated emotional expression, shallow and rapidly shifting emotions, uses physical appearance for attention, inappropriately provocative behavior, impressionistic speech but lacks detail, considers relationships intimate
 - Theme: excessive emotionality
 - Borderline Personality Disorder: a disorder affecting a variety of areas related to relationships, emotions, self, and behavior
 - Symptoms
 - Relationships: unstable intense relationships with alternating idealization and devaluation (splitting), frantic avoidance of abandonment
 - Emotions: intense uncontrolled anger, marked reactivity of mood, chronic feelings of emptiness
 - Self: unstable self-image
 - Behaviors: self-damaging impulsivity, suicidal gestures/self-mutilation, transient stress-related dissociation/paranoia
 - Treatment: dialectical behavioral therapy (DBT)
 - Emphasis on self-acceptance with the need to change; incorporates mindfulness, distress tolerance, emotion regulation, and interpersonal effectiveness with the goal to become aware of the thoughts and emotions that precede destructive behavior with the goal of breaking the sequence and regulating emotions without acting on them
 - Theme: instability
 - Narcissistic Personality Disorder (NPD): grandiose sense of self-importance, preoccupied with success, requires admiration, sense of entitlement, interpersonally exploitative, lacks empathy with haughty attitudes
 - Notes: a narcissistic injury (blow or threat to self-esteem) may result in a rage reaction
 - Theme: grandiosity
 - Antisocial Personality Disorder (ASPD): repeated unlawful acts, deceitfulness, irritability and aggressiveness, reckless disregard for the safety of self or others, irresponsibility (i.e. employment, financial, etc.), lack of remorse
 - Criteria: must be 18 years old or older, there must be evidence of Conduct Disorder with onset before 15 years old
 - Theme: pervasive disregard for others' rights
- Cluster C (anxious, fearful)

- **Avoidant Personality Disorder**: avoids interpersonal and occupational activities for fear of criticism, unwilling to start relationship unless certain of being liked, restraint within relationship for fear of ridicule, reluctance to engage in new activities, negative self image
 - Theme: **feelings of inadequacy**
- **Dependent Personality Disorder**: indecisive, others must take responsibility, difficulty disagreeing, difficulty initiating due to low confidence, excessive lengths to keep/gain support, feels helpless when alone, urgently seeks another relationship if one ends
 - Theme: **excessive need to be cared for**
- **Obsessive-Compulsive Personality Disorder**: preoccupation with details, rules, lists until point of activity is lost, perfectionism interferes with task completion, excessive devotion to work, rigid and stubborn, overly conscientious/scrupulous, reluctance to delegate or to work with others, hoarding and miserly behavior
 - Theme: **inflexible perfectionism & control**
- Treatment: **CBT**

Sleep-Wake Disorders

- To diagnose, interviews are conducted along with polysomnography (PSG) tests that involves measuring a variety of physiological parameters including brain waves, muscle contractions, breathing, etc. during sleep
- Disorders
 - **Insomnia Disorder**: difficulty initiating or maintaining sleep
 - Etiology: a learned arousal response to bedtime and other cues associated with the sleep environment (i.e. inappropriate sleep hygiene)
 - Treatment: improve sleep hygiene, sedatives (**benzodiazepines**), **non-benzodiazepine hypnotics** (fewer side effects)
 - **Benzodiazepines** = use short-term only due to long-term side effects (**decreased SWS - N3 & REM & increased N2**)
 - **Hypersomnolence Disorder**: excessive sleepiness despite sufficient sleep (7+ hours)
 - Symptoms: average sleep episodes of 9.5+ hours, unrefreshing naps, normal PSG
 - Treatment: **stimulants/stimulant-like drugs**
 - **Narcolepsy**: recurrent irresistible sleep occurring within the same day, several times per week
 - Criteria: at least 1+ of the following - cataplexy, characteristic PSG abnormalities (nocturnal PSG - sleep-onset REM; daytime PSG - short REM-sleep latency/sleep-onset), hypocretin deficiency
 - **Cataplexy**: sudden loss of muscle tone while awake that is typically precipitated by strong emotion; considered a REM-related behavior occurring outside of REM sleep
 - Treatment: **stimulants/stimulant-like drugs**, **antidepressants**, **sodium oxybate/GHB** (treats cataplexy and somnolence; **typically schedule I drug but is schedule III when marketed for sleep - tightly regulated**)
 - Breathing-Related Sleep Disorders
 - **Obstructive Sleep Apnea Hypopnea (OSAH)**: multiple episodes of breathing cessation/reduction occur per night due to an **upper airway obstruction** (usually occurs when the soft tissue in the back of the throat collapses during sleep)
 - Sleep-stage arousal: the rise of CO₂ during apnea causes temporary arousal (not awakening) from sleep, which bumps the person from a deep (N3) to a light (N2) stage of sleep; duration of sleep may be adequate but unrefreshing
 - Treatment: **CPAP machine** (device that maintains an open airway by delivering compressed air at a specific air pressure to the mask's nasal pillow), weight loss, avoiding back sleeping, orthodontic devices, surgery
 - **Central Sleep Apnea (CSA)**: multiple episodes of cessation of breathing per night caused by **CNS dysregulation of breathing**
 - Notes: PSG can distinguish OSAH and CSA based on whether thoracic movements occur at the start of an apneic episode (**OSAH - thoracic effort occurs; CSA - thoracic effort does not occur**)
 - Treatment: **respiratory stimulants**, **nocturnal oxygen**

- **Circadian Rhythm Sleep-Wake Disorder (CRSWD)**: excessive sleepiness or insomnia resulting from a mismatch between a person's circadian sleep-wake pattern and the sleep-wake schedule required by the environment
 - Treatment: **phototherapy** to adjust timing of sleep-wake cycle
 - Light → SCN → inhibits pineal gland → decreases melatonin → alert
 - No light → SCN → activates pineal gland → increases melatonin → drowsy
- **Parasomnias**
 - **Non-REM Sleep Arousal Disorder (NRS-AD)**: repeated episodes of incomplete awakening from sleep with either sleepwalking or sleep terrors
 - **Sleepwalking (somnambulism)**: rising from bed and walking about with a blank and staring face, relative unresponsiveness, and difficulty awakening
 - **Sleep terrors**: abrupt terror arousals (usually with panicky scream), intense fear and autonomic arousal, and unresponsiveness to comforting by others
 - Characteristics: episodes occurring within first 1/3 of sleep (during SWS), difficulty awakening/orienting, no/little dream imagery, amnesia for episode the next morning
 - Treatment: **benzodiazepines** (decreases N3 SWS)
 - **Nightmare Disorder**: extremely dysphoric dreams that typically involve threats to survival, security, or physical integrity
 - Characteristics: awakening in 2nd half of sleep period (during REM), rapid alertness upon awakening, dream content is well-remembered, good recall of the awakening the next morning
 - Treatment: **antidepressants** (decreases REM sleep)
 - **REM Sleep Behavior Disorder (RSBD)**: vocalizations and/or complex motor movements occur during REM sleep
 - Characteristics: typically action-filled, violent dreams (**REM sleep without atonia**), immediate awakening, orientation, and alertness with detailed dream recall
 - Treatment: **benzodiazepines (clonazepam)**
 - **Restless Leg Syndrome (RLS)**: urge to move legs in response to uncomfortable sensations with all the following features; patient is aware of symptoms and complains of insomnia
 - Characteristics: occurs/worsens during inactivity, nocturnal worsening of symptoms, temporary relief from discomfort by movement
 - Treatment: **same drugs as for Parkinson's Disease** (also **benzodiazepines, anticonvulsants, etc.**)
 - **Periodic Limb Movement (PLM)**: repetitive muscle contractions during sleep, usually of the lower limb; patient is unaware of movements and complains of daytime sleepiness
 - Characteristics: associated with multiple sleep stage arousals, PSG confirms
 - Treatment: **similar drugs to RLS**

Addiction Disorders

- **Disorders**
 - **Substance Use Disorder**: a maladaptive pattern of substance use over the past 12 months that leads to impairment in social, physical, or occupational functioning. It is characterized by **having 2 or more of the following 11 symptoms within a 12-month period**
 - **Impaired control**: taking a larger amount than intended, persistent unsuccessful attempts to cut back, time consuming, craving or strong desire for the substance
 - **Social impairment**: failure to fulfill major role obligations, social/interpersonal problems related to the substance use, reduction in important activities
 - **Risky use**: use in physically hazardous situations, use despite having a physical/psychological problem related to the substance
 - **Pharmacological**: tolerance, withdrawal symptoms upon cessation
 - **Substance Dependence**: compulsive use of a substance despite the negative consequences
 - **Psychological dependence**: compulsive substance use without withdrawal and tolerance

- **Physiological dependence**: compulsive substance use with withdrawal and tolerance
- **Intoxication**: the development of physical or psychological symptoms due to the recent ingestion of a substance and its CNS effects
- **Withdrawal**: The development of physical or psychological symptoms after the reduction or cessation of intake of a substance
- **Tolerance**: A need for increased amounts of substance to achieve the same physical and psychological effects
- **Gambling Disorder**: persistent and recurrent problematic gambling behavior leading to clinically significant impairment or distress as indicated by the individual experiencing 4+ of the following 9 symptoms in a 12-month period (not due to manic disorder; related to impulse control)
 - Features: preoccupation with gambling, gambling increasing amounts/items of value, unable to control/cut back/stop, used as an escape from problems, gambling to recoup losses, lying to conceal extent of gambling, illegal acts to finance gambling, jeopardizing or losing relationships, reliance on others to pay debts
 - Treatment: **SSRIs and opioid antagonists** (control gambling cravings), cognitive-behavioral therapy (understand triggers and relaxation techniques to manage cravings)
- Types of Substance Abuse
 - **Sedatives**: alcohol, barbiturates, benzodiazepines, inhalants
 - Common features of intoxication: sedation, sleepiness, decrease anxiety, disinhibition, impaired judgment, slurred speech, incoordination, stupor, coma, **respiratory depression** → overdose can be potentially fatal
 - Other potential features: **decreased neuronal firing**, alcohol-related brain damage (i.e. Wernicke-Korsakoff), cross-tolerance to other sedatives
 - Common features of withdrawal: agitation, insomnia, anxiety, autonomic nervous system hyperactivity (tachycardia, hypertension), nausea, vomiting, hand tremors, seizures, hallucinations (any sensory modality, **formication** [sensation of bugs crawling on skin], **alcohol hallucinosis** [main symptom of withdrawal without physical symptoms])
 - Other potential features: **delirium tremens** (confusional state that may also occur as a part of sedative withdrawal; commonly seen after chronic heavy alcohol use)
 - Alcohol
 - Treatment: benzodiazepines (chlordiazepoxide, lorazepam, **diazepam**), competitive inhibitor/antagonist of opioids (**naltrexone** → **reduces the pleasurable effects of alcohol**), unknown mechanism of action (**acamprosate** → NMDA receptor antagonist, **reduces the anhedonia** seen with prolonged abstinence), producing uncomfortable side effects (**disulfiram** → inhibits alcohol dehydrogenase, **alcohol aversion agent**, **only used short-term** and is used to increase motivation to remain abstinent)
 - Screening (**CAGE test**): 2 or more "yes" answers → clinically significant, follow up
 - Have you ever felt you should **cut** down on your drinking?
 - Have people **annoyed** you by criticizing your drinking?
 - Have you ever felt bad or **guilty** about your drinking?
 - Have you ever had a drink first thing in the morning (**eye-opener**)?
 - Barbiturates: phenobarbital, secobarbital
 - Benzodiazepines: diazepam, lorazepam
 - Treatment: **flumazenil** (competitive antagonist, high affinity for benzodiazepine binding site on GABA_A receptor → **used in overdose/toxicity**)
 - Inhalants: paint, glue, etc.
 - Signs: rash, red/runny nose, chemical smell, face discoloration; associated with organ failure and death (**Sudden Sniffing Syndrome**)
- Stimulants: cocaine, amphetamines, ecstasy, bath salts, designer K2, caffeine, nicotine
 - Common features of intoxication: euphoria, grandiosity, psychomotor acceleration and stereotypies, paranoia, hallucinations, loss of appetite, insomnia, chest pains, seizures, **mydriasis**, tachycardia, hypertension

- Features mimics symptoms of other illnesses (mania +/- psychotic features): schizophrenia, bipolar I episodes (both disorders may be clinically indistinguishable from stimulant use)
- Common features of withdrawal: dysphoric mood (mandatory symptom), fatigue, psychomotor slowing, increased appetite, hypersomnia + vivid/unpleasant dreams; non-life threatening
 - Features mimics symptoms of other illnesses (MDD with atypical features)
- Cocaine: similar effects to methamphetamines but shorter-lasting → more-frequently used
- Methamphetamine: meth mouth (dry mouth, dental caries), meth face (repetitive motor movements → picking at skin on face and hands)
- Ecstasy (MDMA (methylenedioxymethamphetamine)/Molly): party drug, stimulant with mild hallucinogenic effects
 - Psychological effects: increased empathy, perceptual changes (things seem more interesting, time and sensory distortion)
 - Other features: increased thirst, hyperthermia, neurotoxicity
- Bath salts: designer drug containing amphetamine-like substances (symptoms more severe than cocaine intoxication with a longer duration)
 - Acute toxicity: agitation, paranoia, hallucinations, chest pain, tachycardia, hypertension, suicidality
- Designer drug K2: synthetic cannabinoid (K2/Spice/Joker/Black Mamba/Kush/Kronic)
 - Acute toxicity: perceptual disturbances, agitation, seizures
- Caffeine: not recognized as DSM disorder
 - Intoxication (similar to hypomania): typically after a dose higher than 250mg of caffeine; symptoms of increased energy, insomnia, nervousness, rambling thoughts, tachycardia, diuresis (blocks adenosine receptors), GI disturbance
 - Withdrawal: headache, dysphoria, fatigue, decreased concentration
- Nicotine:
 - Intoxication: not recognized in DSM
 - Withdrawal: dysphoric mood, restless, anxious, difficulties concentrating, irritable, angry, increased appetite, decreased heart rate
 - Treatment: nicotine replacement therapy (low doses of nicotine to decrease cravings), bupropion (contraindicated in individuals with bulimia nervosa) and varenicline (both increase levels of dopamine)
- Hallucinogens & related substances: cannabis is only substance with withdrawal symptoms
 - Hallucinogens: LSD, mescaline, psilocybin → hallucinations
 - Notes: high potency (effects seen at 25 micrograms), no known overdose deaths, effects last 8-12 hours, no known withdrawal symptoms, low addiction rates
 - LSD may play a role in precipitating onset of acute psychosis in individuals with familial risk of schizophrenia
 - Intoxication: increased heart rate, increased blood pressure, sweating, mydriasis, dehydration, euphoria, paranoia, sensory distortion, visual hallucinations, reduced appetite, wakefulness, time distortions
 - Hallucinogen Persisting Perception Disorder (HPPD): flashbacks of visual hallucinations/distortions experienced during a previous hallucinogenic drug experience
 - Cannabis (sativa, indica, ruderalis): marijuana → distortions
 - 4th most recreationally used substance worldwide (after alcohol, caffeine, and tobacco)
 - Triphasic psychoactive effects: relaxation and slight euphoria, introspection and metacognition (adverse: anxiety and paranoia), increased heart rate and appetite
 - CB1/CB2 cannabinoid receptors mediate the effects of cannabinoids (lasts 2-4 hours, if smoked)
 - CB1 → mostly in the brain

- **CB2** → peripherally throughout the body (modulate pain and inflammation)
 - Notes: low addiction potential, bad trips produce intense negative emotions
 - Intoxication: conjunctival injection, dry mouth, increased appetite, poor muscle coordination, delayed reaction times
 - Withdrawal: irritability, nervousness, dysphoric mood, sleep disturbances, decreased appetite, headaches, night sweats, stomach cramps, shakiness
- Dissociative anesthetics: ketamine, phencyclidine (PCP) → **depersonalization**
 - Intoxication: depersonalization, decreased responsiveness to pain, increased heart rate, increased blood pressure, increased respirations, agitation, belligerent, confusion, impulsivity, unpredictability, **nystagmus**, hyperacusis, ataxia, **muscle rigidity**, seizures, coma
 - Treatment: **benzodiazepines/antipsychotics**, reduce environmental stimulation
 - Withdrawal: **no recognized withdrawal symptoms**
- General side effects: negative experiences (bad trips) produce intense negative emotions → **irrational fears**, anxiety, panic attacks, paranoia, rapid mood swings, hopelessness, **intrusive thoughts** (harming others), **suicidal ideation**
 - No predictive factors for bad trips
- Treatment
 - Research: **use in intractable trauma/stress and anxiety disorders** (breakdown of "Ego" by introducing impermanence of reality and a separation of the self from thoughts and feelings)
- Opioids: heroin, morphine, codeine, methadone, oxycodone, fentanyl
 - Analgesics to reduce pain
 - Intoxication: euphoria, decreased respirations, low blood pressure, constipation, drowsiness, impaired cognitive function, unconsciousness, **miosis**
 - Withdrawal: dysphoria, nausea, vomiting, diarrhea, muscle aches, **lacrimation**, **rhinorrhea**, piloerection, sweating, fever, yawning, **mydriasis**
 - Non-life threatening but very uncomfortable; overdose is deadly
 - Treatment: **naloxone/narcan** (short-acting opioid receptor antagonist, **used for acute overdose** - not addiction → **reverses respiratory and CNS depression**), **naltrexone** (long-acting opioid receptor antagonist that **blocks opioid effects if relapse occurs** → person feels physically unwell), **methadone** (full opioid agonist → **addictive** and potentially dangerous; administered, not prescribed), **buprenorphine** (partial opioid agonist, **less addictive/dangerous**; buprenorphine + naloxone → **naloxone counteracts buprenorphine if abused**)
 - **Protracted Abstinence Syndrome**: anhedonia from downregulation of dopamine and opioid receptors
- Other substances of abuse
 - Anabolic steroids: used as performance-enhancing agents to increase muscle mass and strength
 - Intoxication: acne, masculinization of females, hepatic dysfunction, increased risk of MI, increased libido, aggression
 - Withdrawal: fatigue, depressed mood
- Drug Rehabilitation and Treatment
 - **Acute treatment phase**: in-patient, maximal environmental structure; detoxification and management of withdrawal symptoms/cravings, treat associated medical problems, plan to address comorbid psychiatric illnesses
 - **Recovery phase**: medium environmental structure with the goal to prevent relapse and avoid the Abstinence Violation Effect (feeling bad for relapsing)
- Drug Schedules
 - Schedule I: cannot be prescribed (heroin, marijuana, LSD, ecstasy)
 - Schedule II: can be prescribed but has potential for severe psychological and physical dependence (cocaine, methamphetamine, methadone, oxycodone, fentanyl)

- Schedule III: can be prescribed (ketamine, anabolic steroids)
- Schedule IV: can be prescribed (diazepam)
- Schedule V: can be prescribed (robitussin)

Memory

- Declarative (explicit): facts, events
 - Episodic memory
- Procedural (implicit): skills/habits, conditioning, priming
- Stages of memory: sensory input → sensory memory → encoding → short-term (working) memory + rehearsal ↔ consolidation/retrieval ↔ long-term memory
- Amnesias
 - **Infantile amnesia**: early childhood events cannot be recalled
 - **Transient global amnesia**: occurs typically in older men; recent events and information can only be remembered for a few minutes - normally all other functions are not impaired
 - **Dissociative/psychogenic amnesia**: psychological reaction to severe trauma; memory loss for autobiographical information which doesn't occur due to another disorder
 - **Localized**: total loss of personal memory during a circumscribed period
 - **Selective**: some (but limited) recall of personal memories during a circumscribed period
 - **Generalized**: loss of personal memory of entire life up to and including the event
 - **Wernicke-Korsakoff (diencephalic) amnesia**: caused by thiamine deficiency in patients with alcohol abuse; symptoms include confusion, confabulation, and severe memory impairment
 - **Intact**: IQ, procedural learning, **consolidation**
 - **Deficits**: anterograde memory, retrograde memory, impaired encoding, **poor insight**, **confabulation**
 - **Henry Molaison (bilateral mesial temporal) amnesia**: heavy anterograde amnesia as well as temporally graded retrograde amnesia
 - **Intact**: IQ, procedural learning, **insight**
 - **Deficits**: anterograde memory, retrograde memory, impaired encoding, **impaired consolidation**
- **Engram**: a hypothetical change in neural tissue that is believed to account for the persistence of memory; the connections between neurons are strengthened through repeated co-firing; multi-modal binding of engrams → learning (i.e. seeing a circle + hearing the word "circle" aloud)
 - **Short-term synaptic plasticity**: transient changes in synaptic function (i.e. altered metabolism)
 - **Long-term synaptic plasticity**: more permanent alteration of cellular function (i.e. change in protein expression or cellular structure)
- **Reconsolidation**: reactivation of memory traces renders them labile via involvement of the hippocampal system

Sex

- Sexual Dimorphism: endpoint consists of 2 forms - 1 form in males, 1 form in females (i.e. copulatory behavior, bird song, etc.)
- Sex Differences: endpoint exists on a continuum and average is different between males and females (i.e. pain thresholds, fear, anxiety, learning, memory, etc.)
 - Genetic: every neuron, glia, or other cell type carries either the complement of male chromosomes (XY) or female chromosomes (XX), but not both
 - Hormonal: differential effects of ovarian and testicular hormones
 - Sex Ratios in CNS Disorders: higher frequency of developmental onset disorders in males and adult-onset disorders in females
 - Cognitive Differences: males have a visual rotation advantage while females have an advantage when it comes to language, reading, and verbal fluency
- Sex Convergence: endpoint is the same but neural underpinnings are different (i.e. parental behavior, problem-solving strategies, etc.)

Sleep

- Electroencephalography (EEG): records very small electrical fields generated by synaptic currents in cortical pyramidal cells (signals are low in amplitude so a differential amplifier is used; thousands of neurons must be summed to create a discernible EEG signal)
 - **Desynchronized**: cortical activity causes low (voltage) amplitude EEG waves; normal awake brain
 - **Synchronized**: cortical activity causes high (voltage) amplitude EEG waves; sleeping brain
- Sleep stages: 90 minute cycles, around 7 hours per night, 80% in NREM sleep
 - Awake: alpha & beta waves
 - REM sleep: beta waves; becomes more prominent towards the morning
 - Stage 1 NREM: theta waves
 - Stage 2 NREM: spindles & K-complexes
 - Stage 3 NREM: delta waves; SWS
 - Stage 4 NREM: delta waves; SWS
- Sleep pressure: homeostatic process that keeps track of how long we have been awake; the longer we are awake, the greater the buildup of sleep pressure, possibly reflected by adenosine levels - interacts with circadian sleep rhythm
- Sleep time
 - Babies: sleep for 16 hours/day; spend 50% in REM sleep
 - Adults (20-50 years old): spend 20% in REM sleep
 - Elderly: sleep for 6 hours/day & increased **sleep fragmentation** (due to reduction of galanin-containing neurons in the VLPO)

Sleep Stage	EEG	Sensation	Muscle Tone	Eye Movements
Awake	Low voltage/high frequency; desynchronized	Vivid, externally generated	Continuous voluntary movements	Often
Stages N1-N3	High voltage/low frequency; synchronized	Dull, absent	Occasional involuntary movements	Rare
REM	Low voltage/high frequency; desynchronized	Vivid, internally generated	Muscle paralysis, LMNs inhibited	Often

Drugs

Type	Examples	Notes
Antidepressants		
Selective Serotonin Reuptake Inhibitors (SSRIs)	Fluoxetine (Bulimia Nervosa), Paroxetine, Sertraline, Citalopram	Most popular, benign side effects (GI disturbances, sexual dysfunction)
Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs)	Venlafaxine, Duloxetine	Similar side effects of SSRIs but also increases BP
Tricyclic Antidepressants (TCAs)	Imipramine, Amitriptyline	Blocks presynaptic reuptake of NE & 5-HT; less serotonin agonism; has cardiac-related side effects (orthostatic hypotension)
Monoamine Oxidase Inhibitors (MAOIs)	Phenelzine, Selegiline	Less widely used; diet restrictions to avoid tyramine-induced

		hypertension (i.e. cheese effect)
Sedative-Related (sedatives/hypnotics/anxiolytics)		
Benzodiazepines	Diazepam, Temazepam (short-term use only), Clonazepam (reduces motor movements), Lorazepam, Chlordiazepoxide	
Sleep-Promoting Non-Benzodiazepines	Zolpidem (long-term use approved)	
Barbiturates	Phenobarbital, Secobarbital	
Other Sedatives	Sodium Oxybate	
Mood Stabilizers		
Mood Stabilizers	Lithium, Valproate/Valproic Acid (anti-convulsant); Olanzapine (atypical antipsychotic)	Target manic phase of BPI & depressive phase of BPII/cyclothymia; common side effects include hand tremor/motor disturbances, polyuria/polydipsia; antidepressants contraindicated due to mania
Antipsychotics (neuroleptics)		
Traditional (1st generation) - D2 Receptor Antagonists	Chlorpromazine, Haloperidol	
Atypical (2nd generation) - D2, 3, 4 Receptor Antagonists & 5-HT2A Receptor Antagonists	Clozapine, Risperidone, Olanzapine (atypical antipsychotic)	
Alzheimer's Disease Treatment		
Cholinesterase Inhibitors	Donepezil, Rivastigmine, Galantamine	
NMDA Receptor Blockers	Memantine	
Antibody Drug	Aducanumab	
Stimulants		
Stimulants	Methylphenidate, Amphetamine, Dextroamphetamine, Methamphetamine, Cocaine, MDMA (Ecstasy)	
Norepinephrine Reuptake Inhibitors	Atomoxetine, Viloxazine	
Wake-Promoting Drugs	Modafinil	
Opioids		
Opioids	Heroin, Morphine, Codeine, Oxycodone, Fentanyl, Methadone	

Hallucinogens		
Hallucinogens	Lysergic Acid Diethylamide (LSD), Mescaline, Psilocybin	
Cannabis	Marijuana, Hashish	
Dissociative Anesthetics	Phencyclidine (PCP), Ketamine, Esketamine (antidepressant - blocks NMDA receptors)	
Substance-Abuse Treatments		
Alcohol Use Disorder	Naltrexone, Disulfiram, Acamprosate	
Benzodiazepine Overdose	Flumazenil (GABA antagonist)	
Nicotine Use Disorder	Varenicline, Bupropion	
Opioid Overdose	Naloxone	
Opioid Use Disorder (Abstinence Therapy)	Naltrexone	
Opioid Use Disorder (Replacement Therapy)	Methadone, Buprenorphine	
Other		
Respiratory Stimulants	Acetazolamide (Central Sleep Apnea)	
Anti-Androgens	Medroxyprogesterone Acetate (Paraphilias)	

Other Therapies

- Brain Stimulation Therapies
 - **Electroconvulsive Therapy (ECT)**: electrical induction of a generalized seizure
 - Mechanism: multiple indiscriminate CNS effects
 - Indications: treatment-resistant depression/mania or severe depression/mania
 - Contraindications: vascular disease (due to blood pressure increases)
 - Procedure: frontotemporal cranial placement (unilateral → less effective, fewer side effects; bilateral → more effective, more side effects); creates generalized seizures lasting for 30+ seconds (x3 treatments/week for 6-12 treatments)
 - Side effects: headaches, memory loss for recent long-term memory, memory consolidation difficulties surrounding treatments; normally, memory abilities normalize after a few months post-treatment
 - Efficacy: rapid clinical improvement of mood episode and very safe; used to treat acute illnesses, not as a prophylactic
 - **Repetitive Transcranial Magnetic Stimulation (rTMS)**: magnetic pulses are generated and used to stimulate the prefrontal cortex
 - Indications: patients who have failed at least 1 medication trial
 - Procedure: 40-minute sessions at x5 sessions/week for 4-6 weeks until therapeutic benefits can be seen
 - Side effects: minimal (i.e. scalp discomfort at stimulation site)

- **Vagus Nerve Stimulation (VNS)**: intermittent electrical stimulation of the vagus nerve (CN X) via an implanted wire and pacemaker through surgical intervention
 - Indications: treatment-resistant depression
 - Procedure: surgery to implant wire and pacemaker - comes with typical risks of surgery; therapeutic effect may take several months to be seen
 - Side effects: mild effects of stimulation (i.e. coughing, hoarseness)
- Experimental Therapies
 - **Magnetic Seizure Therapy (MST)**: focal seizure induction
 - Procedure: similar to ECT treatment & rTMS treatment, except seizures are induced
 - Goal: retain effectiveness of ECT but reduce the memory-related side-effects
 - **Deep Brain Stimulation (DBS)**: stimulation of various target sites deep within the brain (i.e. subcallosal cingulate white matter)
 - Indications: OCD treatment

Neurology

Limbic Lobe

- Phylogenetic developments: "lower" animals (i.e. rabbits) have a more well-developed olfactory lobe when compared to humans or even chimpanzees → gets replaced by increased development of the frontal & parietal lobes
- **Papez Circuit**: cingulate gyrus → hippocampus → fornix → mammillary bodies → cingulate gyrus; "Experience & Expression"
 - **Paul MacClean** → added septum and amygdala into the circuit; argued the system was automatic, preverbal, unconscious, and inaccessible to the intellect
- **Low road**: low-resolution visual information goes straight to the amygdala (bypasses V1 - primary visual cortex) → hypothalamus → fight-or-flight
 - Walking in dark, think you see something but can't tell → fear (survival evolution, cortex is slower)
- **High road**: high-resolution visual information goes to the cortex (slower); V1 → amygdala → hypothalamus → fight-or-flight

Theorist	Advance
Broca	- Comparative neuroanatomy of the limbic lobe - Structures: subcallosal gyrus, cingulate gyrus, isthmus, parahippocampal gyrus, uncus
Papez	- Role of limbic circuit in emotion - Cingulate as the main structure supporting emotional experience - Circuit: hippocampus → fornix → hypothalamus → fornix → mammillary bodies → mamillothalamic tract → anterior nucleus of the thalamus → internal capsule → cingulate gyrus → cingulum → entorhinal → hippocampus
MacClean	- Limbic system; Paleomammalian brain - Structures: hippocampus, thalamus, hypothalamus, cingulate, amygdala, septal region
LeDoux	- Low road (SC → pulvinar → amygdala → hypothalamus): fast, automatic, low-resolution, "where" stream, bottom-up - High road (LGN → V1 → corticocortical → amygdala): slow, controlled, high-resolution, "what" stream, top-down

- Lesions
 - **Phineas Gage Case**
 - Lesion: prefrontal cortex
 - Demonstrated that ventromedial frontal lobes play a critical role in behavioral monitoring and emotion regulation

- Symptoms: emotional disinhibition, emotional flattening, increased docility, decreased goal-oriented behavior
- **Kluver-Bucy Syndrome**
 - Lesion: **bilateral temporal** (amygdala, hippocampus, anterior temporal lobe)
 - Symptoms: docility, loss of fear, hyperorality, hyperphagia, hypersexuality, psychic blindness; additional symptoms in humans: memory impairment, aphasia, utilization behaviors, seizures
- Brain Regions
 - **Hippocampus**: responsible for forming context to fear learning; should track the context in which a fear response is learned and help to suppress fear when in safe contexts (and facilitate fear when in unsafe contexts)
 - **Damage**: overgeneralization of fears (think: stimulus generalization)
 - **Amygdala**: responsible for fear learning; initiates fear response via activation of the hypothalamic-pituitary-adrenal (HPA) axis and the sympathetic nervous system
 - **Damage**: inability to form new fear memories
 - **Orbitofrontal cortex**: responsible for long-term fear memory
 - **Ventromedial prefrontal cortex (VMPFC)**: responsible for down-regulation of fear/extinction learning; should inhibit the amygdala when the fear response is no longer adaptive/necessary (important for "extinction memory")
 - **Damage**: inability to form new extinction memory
 - **Hypothalamus**: responsible for the fight-or-flight response
- Fear Conditioning
 - **Conditioning**: formed through classical conditioning
 - Memory network
 - Auditory (tone): tone → auditory thalamus → auditory cortex → basolateral amygdala → central amygdala → central gray (freezing of motor system) + lateral hypothalamus (sympathetic nervous system activation) + bed nucleus of stria terminalis (hypothalamus-pituitary-adrenal axis)
 - Somatosensory (shock): shock → somatosensory thalamus → somatosensory cortex → basolateral amygdala → central amygdala → central gray (freezing of motor system) + lateral hypothalamus (sympathetic nervous system activation) + bed nucleus of stria terminalis (hypothalamus-pituitary-adrenal axis)
 - Fear learning: **ventromedial prefrontal cortex is suppressed, amygdala is active**
 - **Extinction**: formed through repeated disassociation of US + CS to remove CR
 - Memory network
 - Auditory (tone): tone → auditory thalamus → auditory cortex → ventromedial prefrontal cortex (VMPFC) → amygdala
 - Somatosensory (no shock): no shock → somatosensory thalamus → somatosensory cortex → ventromedial prefrontal cortex (VMPFC) → amygdala
 - Fear learning: **ventromedial prefrontal cortex is active, amygdala is suppressed**
 - **Contextual**: formed when a US is paired with a specific neutral stimulus + CS in order to discriminate CR responses (i.e. blue cage = shock with tone; red cage = no shock with tone)
 - Memory network
 - **Unsafe environment** (i.e. blue cage)
 - Auditory (tone): tone → auditory thalamus → auditory cortex → hippocampus → excitatory response to → amygdala
 - Somatosensory (shock): shock → somatosensory thalamus → somatosensory cortex → hippocampus → excitatory response to → amygdala
 - **Safe environment** (i.e. red cage)
 - Auditory (tone): tone → auditory thalamus → auditory cortex → hippocampus → inhibitory response to → amygdala

- Somatosensory (no shock): no shock → somatosensory thalamus → somatosensory cortex → hippocampus → inhibitory response to → amygdala
- Fear learning in **unsafe environment**: hippocampus up-regulates the amygdala
- Fear learning in **safe environment**: hippocampus down-regulates the amygdala

Temporal Lobe

- **Temporal lobe epilepsy**
 - Common symptoms: aura (deja-vu, sense of fear, olfactory/gustatory sensations, rising epigastric sensations), behavioral automatisms (lip smacking, automatic movements), attention/cognitive comorbidities, ictal amnesia (amnesia post-seizure), mood changes (anxiety, depression)
 - **Research correlate**: children with epilepsy and comorbid anxiety disorder have larger amygdalas and smaller prefrontal cortexes relative to children with epilepsy but no anxiety disorder

Circadian Dysfunction

- **Jet Lag (Desynchronization)**: transient disorder associated with excessive daytime sleepiness, sleep-onset insomnia, and frequent arousals from sleep
 - Suprachiasmatic nucleus (SCN) resets to local time around 1 day later based on light signals, but peripheral oscillators (liver, pancreas, muscle, adrenal glands, vasculature, adipose tissue) can take more than a week to adjust
 - **Phase advance**: going to bed earlier than the circadian bedtime (i.e. before the production of melatonin)
 - **Phase delay**: going to bed later than the circadian bedtime (i.e. after the production of melatonin); **humans can tolerate a phase delay better than a phase advance** (i.e. it is better to travel from Paris → NY than it is to travel from NY → Paris)
- **"Social" Jet Lag**: differences in an individual's chronobiology (i.e. whether they are larks vs owls)
 - Adolescents and young adults tend to be more "owl-like" (i.e. go to sleep later)
 - Children and older adults tend to be more "lark-like" (i.e. go to sleep earlier)
 - Individual differences may have a genetic component (**Familial Advanced Sleep Phase Syndrome & PER2 gene**)
 - **Night shift work**: associated with increased psychiatric disorders, gastrointestinal disorders, and higher blood pressure
- **Sleep-Wake Disorders in Blind Individuals**
 - Continual circadian desynchrony through a failure of light information to reach the suprachiasmatic nucleus (SCN) resulting in cyclical episodes of poor sleep and daytime dysfunction
 - **Daily melatonin administration** provides a synchronizing daily "time cue" is a promising therapeutic strategy, along with **social rhythm therapy** to set peripheral clocks
 - It is possible that **blue-light sensitive retinal ganglion cells** are functional in some visually-impaired individuals
- **Sleep Deprivation & Sleep Debt**
 - Increased sleep deprivation = increased sleep debt
 - After 2 weeks with 6 or fewer hours of sleep per night, performance falls to the same level as having achieved 24 hours of sleep deprivation
 - **Performance impairment**: poor concentration, distractibility, reduced vigilance, longer reaction time, lack of energy, poor coordination
 - **Mood symptoms**: irritability, lack of motivation, anxiety, depression, poor self-care
 - **Neurologic signs** of sleep deprivation: mild inconstant nystagmus, impaired saccadic eye movements, ptosis of eyelids, slight hand tremor, flat speech, anomia, reduced seizure threshold, perceptual distortions
 - **Health signs** of increased sleep debt: high blood pressure, stroke, heart attack, weight gain, diabetes, immune system suppression
 - **Sleep & Memory**

- Memory encoding takes place during wakefulness, while memory consolidation takes place during sleep
- Synaptic pruning occurs during sleep to enhance memory signals and prune away “noise”
- **Memory replay**: spatiotemporal patterns of neuronal firing that occur during exploration of a novel environment are reactivated in the same order during sleep

Addiction

- Stimulation of reward pathway and overrides punishing effects
 - Neurons in the **ventral tegmental area (VTA)** send their **dopaminergic axons** to the nucleus accumbens, striatum, and the prefrontal cortex → **structures involved in motivation**
 - **Nucleus accumbens (ventral striatum)**: greater release of dopamine seen here → mediates positive reinforcing effect of drugs
- Dysfunction of the prefrontal cortex: stimulation of the reward pathway affects the functioning of the prefrontal cortex and alters self-control
- **Acute withdrawal symptoms**: typical onset within 48-72 hours and lasts 1-2 weeks; discomfort may cause relapse
- **Protracted Abstinence Syndrome**: repeated drug use **decreases the availability of dopamine** as the brain adapts to having drug-induced dopaminergic spikes → associated with prolonged feelings of anhedonia lasting several months which may trigger relapse
- Substance use can impact the frontal cortex: impulse control and inhibition, judgment and decision-making, planning, reward pathway, tolerance, withdrawal, craving
 - Behavior conflict can cause justifications: denial, minimization, defiance, rationalization
 - “Replacement” of the frontal cortex (**proxy executive functions**) followed by resilience-building to life stressors

Consciousness

- Divisions of Consciousness
 - Background Conditions
 - Level of consciousness: arousal → reticular activating system
 - Full Conscious Experience
 - Conscious awareness → higher-order thalamic nuclei, synchronous firing of broad heteromodal cortical network
 - Percept
 - Content of consciousness → specific thalamic relay nuclei, content-specific sensory regions
- Perception: conscious perception relies on both “bottom-up” and “top-down” activation
 - “Top-down” selection is facilitated by long-range bidirectional re-entrant/recursive association fibers connecting prefrontal and parietal regions to sensory regions
 - $V1 \longleftrightarrow V2 \longleftrightarrow V4 \longleftrightarrow$ prefrontal cortex (PFC)
- Levels of Consciousness
 - Lethargic: patient can be fully aroused
 - Obtunded: patient cannot be fully aroused
 - Stuporous: sleep-like status
 - Comatose: no purposeful response to stimuli
- Reticular Activating System (RAS)
 - The major cholinergic nuclei in the RAS are: pedunclopontine nucleus & laterodorsal tegmental nucleus
 - Cholinergic pathways arising from the RAS project to: thalamus (intralaminar nuclei) → cortex & basal forebrain → cortex
 - The brainstem and basal forebrain cholinergic systems work together to abolish cortical slow wave activity and promote an alert state
- Coma States
 - Causes
 - Small lesions in the mesencephalon → paramedian reticular formation

- Lesion of posterior lateral hypothalamus → pathway through hypothalamus
- Lesion of thalamus
- Bilateral impairment to both hemispheres → trauma/metabolic
- Supratentorial mass lesions → epidural/subdural/intracerebral hemorrhages, cerebral infarction, brain tumor, brain abscess
- Subtentorial mass lesions → brain stem infarction/tumor/hemorrhage
- Metabolic and diffuse cerebral disorders → anoxia/ischemia, concussions, ongoing seizures and postictal states, infection, subarachnoid blood, hypo/hyperglycemia, hypo/hyponatremia, hypothyroidism, hypocortisolism, hypercarbia, renal failure, drugs/alcohol, liver failure, sepsis
- Types
 - **Vegetative State** ("Awake Coma"): develops after a coma and results in the loss of ability to think/speak/respond; no awareness of environment
 - Intact: breathing and circulation, autonomic functions, non-cognitive functions, normal sleep patterns, spontaneous emotional expressions
 - **Cause**: bilateral middle cerebral artery (MCA) infarcts
 - **Locked-In Syndrome**: tetraplegia (paralysis of all voluntary muscles with the exception of vertical eye movements)
 - Intact: full awareness of environment, cognition (thinking, remembering, visualizing); EG & cortical metabolic activity are typically normal
 - **Cause**: blockage of the basilar artery leading to a pons infarct
 - **Brain Death**: irreversible loss of all brain functions; not responsive to speech/pain/other stimuli, no spontaneous respiration, dilated pupils unresponsive to light, no vestibulo-ocular reflex or corneal reflex, isoelectric EEG
 - **Cause**: anoxia, ischemia, intracranial hemorrhage, trauma, brain tumors, increased ICP/uncal herniation
- Glasgow Coma Scale
 - 1-15 range; 8 is critical score
 - Less than or equal to 8 → 90% are in coma; at 6 hours → 50% will die
 - Greater than or equal to 9 → not in coma
 - 9-11 → moderate severity
 - 12-15 → minor injury
- Posturing
 - **Decorticate posturing**: damage to bilateral upper midbrain (above red nucleus) → BLE extended, RUE flexed
 - **Decerebrate posturing**: damage to upper pons or lower midbrain (below red nucleus) → BLE + RUE extended; Babinski sign
- Pupillary Light Responses
 - **Metabolic encephalopathy, drug ingestion, or diffuse pressure on the diencephalon**: pupils are slightly smaller than normal but respond to light
 - **Pressure on the pretectal area (pineal tumor)**: no pupillary constriction (large, unreactive pupils)
 - **Lesion of the oculomotor nerve (CN III) via uncal herniation**: unilateral fixed dilated pupil
 - **Lesion in pons**: "pinpoint pupils", responding slightly to light (magnifying lens)
- Testing
 - **Doll's Head Maneuver**: eyes roll counter to head movement → brain stem intact
 - **Caloric Testing**: cold water in ear makes eyes turn to ipsilateral side of water → brain stem intact

Sex

- **Sexually Dimorphic Nucleus (SDN)**: spherical in males (but 4x larger in rats) and more elongated in females (located bilaterally to the 3rd ventricle in both males & females)
- **Interstitial Nuclei of the Anterior Hypothalamus (INAH-3)**: x2 times larger in males than in females in volume, neuronal number, and neuronal size

- Marker for sexual orientation, smaller in homosexual than in heterosexual men, same size in women and homosexual men, smaller in transgender persons (male → female) than in males, similar size in transgender persons (male → female) as in females
- Sexually Dimorphic Areas of the Hypothalamus: interstitial nucleus of the anterior hypothalamus 1/2/3/4 (INAH-1, INAH-2, INAH-3, INAH-4), paraventricular nucleus (PVN), ventromedial nucleus (VMN), supraoptic nucleus (SOP), suprachiasmatic nucleus (SCN), bed nucleus of the stria terminalis (BNST)
 - Bed nucleus of the stria terminalis (BNST): larger in males than in females, smaller in transgender persons (male → female) than in males, similar size in transgender persons (male → female) as in females
- Females have higher gray matter density in the following regions: left/right caudate, hippocampus, gyrus rectus; left inferior frontal lobe, right orbitofrontal lobe
- Sex-Determining Region on Y (SRY) gene: determining factor for sexual differentiation
 - XY + SRY gene → testes
 - XX + no SRY gene → ovaries
 - XX + SRY gene → testes
- Critical Periods
 - 12-20 weeks: sex differentiation (estradiol, progesterone, testosterone)
 - Low estradiol → high estradiol + progesterone (during puberty)
 - High estradiol → high testosterone (during puberty)
 - Puberty: development of behavioral phenotypes (masculinization vs. feminization)
- Androgen Insensitivity Syndrome (AIS): caused by a mutation in the gene that produces androgen receptors
 - XY + SRY gene → testes → female phenotype (body & brain)
- Diethylstilbestrol (DES) in rat experiments
 - Sexually dimorphic nucleus (SDN) larger in males than in females
 - SDN larger in DES-treated female than in female controls (DES also blunted luteinizing hormone secretion)
 - DES exposure during the critical period of sexual differentiation of the brain causes masculinization of the brain

Sleep

- Wakefulness systems: cholinergic (projects to thalamus + cerebral cortex) & monoaminergic (projects directly to cortex) pathways
 - Ascending reticular activating system (ARAS)
 - Pedunculopontine nuclei (PPT), laterodorsal tegmental nuclei (LDT), raphe nuclei (RN; serotonergic), locus coeruleus (LC; noradrenergic), ventral tegmentum (VT; dopaminergic)
 - Hypothalamus
 - Lateral hypothalamus (LH), tuberomammillary bodies (TMN; histaminergic)
 - Basal forebrain (BF)
 - Septal nuclei, nucleus basalis
- Sleep-promoting systems: GABAergic pathways (galanin neurons)
 - Ventrolateral preoptic region (VLPO) of the anterior hypothalamus turns off the arousal systems in the ascending reticular activating system (ARAS)
- Midbrain is the key region of wakefulness (lesion at midbrain leads to decreased responsiveness)
- Thalamus contains sensory relay stations that are in "transmission mode" during the waking state but go into "burst mode" during slow-wave sleep (SWS)
 - Lateral geniculate nucleus (LGN) → vision
 - Medial geniculate nucleus (MGN) → audition
 - Ventral posterolateral thalamus (VPL) → somatosensory
- Suprachiasmatic nucleus strongly innervates the subparaventricular zone (SPZ) of the hypothalamus and the dorsomedial hypothalamic nuclei (DMH)
 - DMH integrates feeding, temperature, and light cues from the SPZ and the SCN; it has GABAergic projections to the VLPO and excitatory projections to the lateral hypothalamus (LH)
 - DMH is an important conduit for light and other environmental cues to influence the sleep/wake switch

- REM sleep
 - Increased blood flow in various brain regions when compared to NREM sleep
 - Pons, midbrain, basal ganglia, thalamus, basal forebrain, paralimbic regions (anterior insula, anterior cingulate, mesial temporal), and association cortices (visual, auditory)
 - Paralysis
 - Supraolivary medulla in the ventromedial medulla → glutamate → spinal cord ventral horn interneurons → GABA/Glycine → motor neurons
- Awake state: increased rCBF in various brain regions when compared to REM sleep
 - Heteromodal association cortices and the dorsolateral prefrontal cortex

Biological Rhythms

- Cyclical Activities
 - Ultradian: takes < 1 day to complete a cycle (i.e. respiration, heart rate, etc.)
 - Circadian: takes ~ 1 day to complete a cycle (i.e. sleep-wake cycle)
 - Infradian: takes > 1 day to complete a cycle (i.e. menstrual cycle)
- Respiratory Rhythm Generation
 - Respiration retained if there is a lesion above the medulla
 - Respiration ceases if there is a lesion below the medulla
 - Cortex
 - Cortical controls are responsible for the voluntary control of breath/breathing
 - Hypothalamus
 - Signals the ventral respiratory group; pain and strong emotion influence breathing rate and depth; increased temperature increases breathing rate
 - Pons
 - Pneumotaxic Center (Pontine Respiratory Group): essential for maintaining normal breathing pattern; inhibits apneustic center; signals the ventral respiratory group, fine-tunes breathing rhythm in sleep/speech/exercise
 - Apneustic Center: lower pons - tonic facilitation of inspiration by excitatory input to the pre-Bötzinger complex
 - Medulla
 - Dorsal Respiratory Group (DRG): inputs from the pulmonary stretch receptors, CN IX, CN X, and outputs to the ventral respiratory group; signals the ventral respiratory group, integrates peripheral signals, and receives visceral afferents
 - Ventral Respiratory Group/Column (VRG): a group of nuclei in the ventrolateral medulla
 - Pre-Bötzinger Complex: the respiratory rhythm generator; primary source of rhythmic inspiratory drive
 - Bötzinger Complex: feedback loop with the pre-Bötzinger complex; expiratory
 - Rostral Ventral Respiratory Group: contains inspiratory neurons
 - Caudal Ventral Respiratory Group: contains expiratory neurons
 - Spinal Cord
 - Phrenic Motor Nucleus (PMN): located in the ventral horn of the cervical spinal cord (C3-C5 level)
- Chemical Modulation
 - Increase in respiration & respiratory rate is induced by: decrease in oxygen, decrease in pH, increase in carbon dioxide
 - Decrease in respiration & respiratory rate is induced by: increase in oxygen, increase in pH, decrease in carbon dioxide
 - Carotid Bodies
 - At the fork of the common carotid artery; sends information mainly through the glossopharyngeal nerve (CN IX); is sensitive to pO₂, pCO₂, and pH
 - Aortic Bodies
 - On the aorta; sends sensory information to the medulla through the vagus nerve (CN X); is sensitive to pO₂, pCO₂, but not pH
- Respiration & CNS Disorders

- Lesion to corticospinal/corticobulbar fibers (forebrain) → Cheyne-Stokes breathing (periods of no breathing followed by periods of waxing and waning of respiratory depth; patients may completely lose voluntary control of respiration)
- Lesion to midbrain → hyperventilation
- Lesion to pontine respiratory group → apneusis/apneustic breathing (prolonged inspiratory plateaus of high lung volume followed by prolonged expiratory pauses with low lung volume)
- Lesion to pre-Bötzinger complex + Bötzinger complex (lower pons/upper medulla) → ataxic breathing (randomly occurring large and small breaths alternating with prolonged periods of apnea)
- Lesion to bilateral medulla → respiratory arrest
- Circadian Rhythm
 - Body temperature is highest at midday and lowest during peak sleep
 - Melatonin levels are highest during peak sleep and lowest at mid-day
 - Cortisol levels are highest during the early waking hours and lowest right before bedtime
 - Zeitgeber: regulates the circadian rhythm - without it, the circadian rhythm would be free-running and would gradually shift further in the day (i.e. instead of a day lasting from 0600-2400 it would gradually shift towards lasting from 1200-0600)
 - Suprachiasmatic nucleus (SCN)
 - Located in the anterior hypothalamus bordering the 3rd ventricle (neurons are among the smallest in the CNS)
 - Pathway: light activates photosensitive retinal ganglion cells → retino-hypothalamic tract fibers synapse in the suprachiasmatic nucleus of the hypothalamus and resets the circadian clock → paraventricular nucleus of the hypothalamus → spinal cord (superior cervical ganglion/SCG) → circadian clock signals reach the pineal gland to induce the release of high levels of melatonin during prolonged darkness
 - SCN → paraventricular nucleus of the hypothalamus: pineal gland pathway → melatonin release
 - SCN → dorsomedial nucleus of the hypothalamus: ventrolateral preoptic region (GABA/galanin) and lateral hypothalamus (hypocretin/orexin) → sleep-wake cycle
 - SCN → subparaventricular zone of the hypothalamus: medial preoptic region (hypothalamus) → body temperature
- Circadian Desynchrony (aging, shift work, jetlag, etc.) Pathologies
 - Brain: sleep/wake cycle + desynchrony → sleep disorder
 - Liver: lipogenesis + desynchrony → dyslipidemia
 - Pancreas: insulin secretion + desynchrony → hypoinsulinemia
 - Muscle: glucose disposal + desynchrony → insulin resistance
 - Adrenals: stress response + desynchrony → chronic stress
 - Vasculature: vascular tone + desynchrony → thromboembolic events
 - Adipose: adipogenesis + desynchrony → obesity

Memory

- Hippocampal Trisynaptic Circuit: sensory stimulus (visual, auditory, olfactory) → entorhinal cortex → perforant pathway → dentate gyrus → CA3 pyramidal cells → fornix / → Shaffer collaterals → CA1 pyramidal cells → subiculum → entorhinal cortex
 - Pathway generates synaptic plasticity
- Long-Term Potentiation in the Hippocampus
 - High-frequency tetanus → potentiated response (via large elevation of [Ca²⁺])
 - Moderate-frequency input → no response
 - Pathway
 - Glutamate released from presynaptic terminals → glutamate binds with both AMPA and NMDA receptors on the postsynaptic membrane → binding opens AMPA channels for Na⁺ entry → EPSPs summed to elicit depolarization → voltage-gated opening NMDA receptor channels by removal of Mg²⁺ → Ca²⁺ entry through NMDA channel activates 2nd messenger pathways (Ca²⁺/calmodulin kinase, PKC, tyrosine kinase) → 2nd

messenger pathways activate protein kinases, mRNA modeling of new AMPA receptors →
 2nd messenger pathways trigger release of retrograde messengers → retrograde
 messengers stimulate lasting increase in presynaptic glutamate release

- Increased synaptic strength/efficiency → increased AMPA channel conductance, increase in number of AMPA receptors, increase in the number of synapses involving the formation of new dendritic spines and synaptic boutons
- Postsynaptic neuron strongly depolarized: large amounts of Ca²⁺ enter postsynaptic neuron → activation of protein kinases
- Long-Term Depression in the Hippocampus
 - Low-frequency tetanus → depressed response (via small elevation of [Ca²⁺])
 - Moderate-frequency input → no response
 - Decreased synaptic strength/efficiency → decreased AMPA channel conductance, decrease in number of AMPA receptors, decrease in synaptic density
 - Postsynaptic neuron weakly depolarized: small amounts of Ca²⁺ enter postsynaptic neuron → activation of protein phosphatases (cause dephosphorylation and internalization of AMPA receptors)

Language

- Brain Hemispheres
 - Right hemisphere
 - Communicative and emotional prosody (stress, timing, intonation): right anterior hemisphere damage → wrong intonation; right posterior hemisphere damage → difficulty in interpretation
 - Pragmatics of language: right hemisphere damage → difficulty in construction of sentences into a story, difficulty in understanding jokes/sarcasm
 - Left hemisphere
 - Language comprehension/expression, lexicon, phonetic assembly/processing
- Wernicke-Geschwind Model of Language
 - Broca's area: language expression
 - Wernicke's area: language comprehension
 - Arcuate fasciculus: communication between Broca's & Wernicke's areas; repetition
- Alexia: word blindness, inability to read caused by a disconnection between the visual and language system (vision is bilateral while language is lateralized); results from disruptions in transfer of visual information to the areas of the left hemisphere
 - Cause: damage in the splenium (posterior part of the corpus callosum)
- Dyslexia: difficulty in reading and spelling despite normal eyesight and hearing, adequate education, and normal IQ that is seen in 10-30% of the population
 - Cause: underdeveloped phonological awareness; abnormalities in the ventral (word identification) and dorsal (orthographic-to-phonetic) visual-auditory pathways
- Aphasias

	Broca's Aphasia	Wernicke's Aphasia	Gerstmann Syndrome	Conduction Aphasia	Global Aphasia	Transcortical Sensory Aphasia	Transcortical Motor Aphasia
Location of lesion	Inferior frontal cortex (Brodmann's area 44 & 45)	Posterior sector of left auditory association cortex (posterior superior temporal gyrus; Brodmann's area 22)	Angular gyrus (Brodmann's area 39)	Arcuate fasciculus	Combination of Broca's, Wernicke's, and Conduction aphasias	Occipito-temporal-parietal watershed zone between posterior and middle cerebral artery territories	Frontal watershed zone between middle and anterior cerebral artery territories
Pronunciation speech rhythm	Dysarthria, stuttering, effortful	Normal	Normal	Normal	Abnormal	Normal	Nonfluent, dysarthric, stuttering
Speech content	Missed syllables, agrammatical,	Use of wrong words or	Normal	Some wrong words	Abnormal	Abnormal	Agrammatical

	telegraphic	non-existent words					
Repetition of speech	Abnormal	Abnormal	Normal	Abnormal	Abnormal	Normal	Normal
Comprehension of spoken language	Normal	Abnormal	Normal	Abnormal	Abnormal	Abnormal	Normal
Comprehension of written language	Normal	Abnormal	Abnormal	Normal	Abnormal	Abnormal	Normal
Writing	Clumsy, agrammatical, misspelling	Penmanship is okay but there are misspellings and inaccuracies	Abnormal	Occasional spelling and language areas	Abnormal	Abnormal	Abnormal
Naming	Better than spontaneous speech	Abnormal	Abnormal	Occasional wrong names	Abnormal	Abnormal	Abnormal
Other	Hemiplegia, apraxia	Hemianopia (sometimes)	Slight hemiparesis, hemianopia	Slight hemiparesis, neglect of right-sided stimuli	Hemiplegia, hemiparesis	Hemianopia, hemisensory loss	Hemiplegia, apraxia

Neuroanatomical Correlates to Psychological Disorders

- **Obsessive-Compulsive Disorder (OCD)**
 - Cortico-striato-thalamo-cortical (CSTC) circuit overactivity: cingulate gyrus → striatum → thalamus → orbitofrontal cortex
 - Serotonin deficiency
- **Post-Traumatic Stress Disorder (PTSD) & Acute Stress Disorder (ASD)**
 - Hyperresponsive amygdala (fear)
 - Under-responsive prefrontal cortex (failure to inhibit fear)
 - Reduced volume and dysfunction of hippocampus (impaired memory consolidation → dissociation)
- **Dissociative Identity Disorder (DID)**
 - Smaller volumes of the hippocampus, amygdala, parietal structures (perception and personal awareness), frontal structures (executive functioning)
- **Major Depressive Disorder (MDD)**
 - Increased activity of the amygdala
 - Chronically increased activity of the hypothalamus-pituitary-adrenal (HPA) axis
 - Increased cortisol levels, cytokines increase inflammatory response (relates to sickness behavior)
 - Decreased gray matter volume in the parahippocampal gyrus & hippocampus bilaterally
 - Decreased volume of the hippocampus
 - Decreased activity & volume of the dorsolateral prefrontal cortex
 - Decreased neurotransmission of monoamines
- **Mania (BPI)**
 - Increased monoaminergic activity
- **Schizophrenia (S)**
 - Mesocortical pathway: ventral tegmental area (VTA) → dorsolateral prefrontal cortex (executive functions) + ventromedial prefrontal cortex (emotions & affect)
 - Underactivity of mesocortical dopamine relates to negative symptoms
 - Mesolimbic pathway: ventral tegmental area (VTA) → nucleus accumbens (NA) → basal forebrain (reward, reinforcement, motivation, motor function)
 - Overactivity of mesolimbic dopamine relates to positive symptoms
 - Nigrostriatal pathway (extrapyramidal syndrome): substantia nigra → caudate + putamen [striatum] (stimulation of purposeful movement)

- Tuberoinfundibular pathway (increased prolactin release): hypothalamus → infundibular region (normally inhibits prolactin release)
- Increased dopamine release in the absence of stimuli
- Enlarged lateral ventricles
- Decreased cortical gray matter and reduced hippocampal volume
- Decreased volume of thalamus
- Decreased prefrontal cortex metabolism
- Impairment in various areas related to attention, memory, executive functions, etc.
- **Antisocial Personality Disorder (ASPD)**
 - Associated with a lack of physiological reactivity to stressors → amygdala dysfunction?
- **Alzheimer's Dementia (AD)**
 - Neuroanatomical: cortical atrophy, hippocampal atrophy, enlarged ventricles
 - Neurochemical: multiple neurotransmitter deficiencies (loss of cholinergic acetylcholine neurons in the **nucleus basalis of Meynert** due to its role in memory formation)
 - Neurofunctional: posterior hypometabolism (parietal/temporal lobes)
 - Histopathological: **beta-amyloid plaques and neurofibrillary tangles**
 - In vivo biomarkers: CSF amyloid & tau levels, PET imaging of amyloid plaques
 - Note: **definitive AD diagnosis depends on postmortem histopathological confirmation**